

# Inferring the Aggregate: Information Frictions and Fiscal Policy

Jonas Overhage<sup>†</sup>

IIES, Stockholm University

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## Abstract

Households use their own experiences to form expectations about the macroeconomy. Changes in household income can thus affect decisions through both resources and information. I study this mechanism in a heterogeneous-agent model in which households infer aggregate movements from their own cash flows. When macroeconomic conditions are not fully observed, countercyclical transfers are not ordinary windfalls, as they signal to households that aggregate conditions are weak. This paper studies how important this information channel is for the effects of such transfers. Evidence from a randomized information experiment in Germany supports this mechanism: fiscal information lowers GDP forecasts and reduces durable spending intentions, with the decline in spending strongest among high-income households. To solve the model, I develop a global method that combines Bayesian filtering with approximate aggregation. Information frictions generate excess smoothness and reduce the two-quarter cumulative consumption response from more than 70 cents per initial dollar transferred under full information to about 15 cents. Individual consumption responses differ across the distribution, even becoming negative at the top.

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<sup>†</sup>web: [joverhage.github.io](https://joverhage.github.io); email: [jonas.overhage@iies.su.se](mailto:jonas.overhage@iies.su.se)

# 1 Introduction

Households have imperfect information about the macroeconomy. Changes in household income can reflect both idiosyncratic circumstances and macroeconomic movements, so realized cash flows contain information about aggregate conditions. What households infer from these cash flows matters for their expected future income and spending. Cash flows therefore affect consumption through both resources and information. This paper studies how important this learning is for household spending, particularly in response to countercyclical fiscal transfers.

I study this question in a heterogeneous-agent model, in which households face borrowing constraints and differ in income as well as wealth. I show that, when the full-information assumption typical of heterogeneous-agent models is relaxed, households rationally infer aggregate conditions from their own realized cash flows. On balance, this information friction lowers the consumption response to a fiscal expansion by about 80% and causes it to exhibit excess smoothness, bringing the model closer to empirical evidence.

To motivate the modeling choice of deviating from full information, I turn to a large empirical literature on expectation formation, as well as my own analysis using the New York Fed Survey of Consumer Expectations (SCE). Both my findings and the broader literature document imperfect information and significant extrapolation by households from their own economic conditions to aggregate ones.

Against this background, countercyclical transfers are a natural fiscal policy instrument to study: They are salient, directly observed cash flows that respond to macroeconomic conditions. A countercyclical stimulus payment is then a negative signal about aggregate conditions, and households react to this information by lowering consumption. In general, households incorporate fiscal policy into their expectation formation. However, different policies appear to carry different information. News about procyclical or acyclical fiscal expansions leads to more optimistic short-run expectations (Weber et al., 2022; Ludolph et al., 2026). In contrast, information about countercyclical policy can instead induce pessimism, with households interpreting the policy response as a signal that aggregate conditions are weak (Coibion et al., 2022; Melosi et al., 2025). Such pessimistic revisions could then depress household spending, offsetting part of the stimulus provided by the

policy itself.

A concrete example comes from an information experiment in the Bundesbank Online Panel – Households (BOP-HH), in which households were informed of a EUR 750 billion fiscal package in April 2020. Metiu and Stockerl (2024) already document that this treatment induces pessimistic revisions in expectations. My analysis of this experiment shows that information about the package lowers expected GDP over the next twelve months by 1.76 percentage points and the stated probability of a large durable purchase over the same period by 6.15 percentage points relative to the placebo group. GDP expectations fall broadly, but the decline in spending intentions is strongest among high-income households, consistent with unconstrained households having more scope to adjust spending. These findings concern information about the policy, rather than a transfer receipt. Thus, they speak directly to the information channel central to the model mechanism.

The information mechanism is first demonstrated in a basic partial-equilibrium model. In this model, I show that when households cannot perfectly distinguish aggregate productivity movements from idiosyncratic wage movements, they rationally infer the former from the latter. The information friction has two implications: First, households learn about persistent shocks only gradually, with the response to aggregate shocks exhibiting excess smoothness and materializing over time. Second, because each cash flow provides imperfect information to the household, the consumption response to an income shock now has an information component.

The mechanism is then brought into a quantitative fiscal-policy model featuring heterogeneous agents. The two implications from the basic model carry over, although information is now conveyed differently: If fiscal policy responds to falling output with transfers, the transfers themselves become a signal, with households inferring that aggregate conditions may be weak when receiving them.

The fiscal-policy model features heterogeneous agents, dispersed information, and endogenous aggregate output. Solving such models requires jointly determining individual policies, belief dynamics, and the cross-sectional distribution. Because standard solution methods do not apply to this setting directly, I develop a method combining Bayesian filtering and approximate aggregation to solve approximately for an equilibrium. The approximation enters through a parametric perceived law of motion. Conditional on this

law, households form expectations optimally using Bayesian filtering, and its coefficients are required to accurately match the true, simulated aggregate dynamics of the model.

Quantitatively, information frictions reduce the two-quarter aggregate consumption response from above 70 cents per dollar transferred under full information to approximately 15 cents. The direct cash response is similar across information structures, while the negative news conveyed by the transfer accounts for most of the decrease. Responses also decline with income and become negative at the top, qualitatively matching the experimental evidence from the BOP-HH data.

These results also help account for modest household and aggregate consumption responses to countercyclical stimulus payments (Orchard et al., 2025, 2026).<sup>1</sup> Strong responses to other cash receipts need not carry over to stimulus payments when the latter also convey information about aggregate conditions.

The paper makes three contributions. First, it identifies and empirically disciplines an information channel of fiscal transfers. Second, it introduces learning from endogenous signals into a heterogeneous-agent model. Third, it develops a global method for computing approximate equilibria in this class of models. The remainder of the paper presents the empirical evidence, develops the mechanism, and evaluates its quantitative implications.

## 2 Related literature

Expectations about aggregate variables are widely documented to be heterogeneous, and a large empirical literature links this heterogeneity to households' own experiences. Kuchler and Zafar (2019) show that households extrapolate from locally experienced house-price changes when forming expectations about future house-price growth. Cavallo et al. (2017) and D'Acunto et al. (2021) document that inflation expectations respond to the prices households themselves observe, especially grocery prices. Weber et al. (2022) survey this evidence. Taubinsky et al. (2024) show how these expectations respond to other idiosyncratic shocks such as income. Closest to the mechanism here, Adams and Rojas (2024) find that households cannot separate idiosyncratic from aggregate components of their own

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<sup>1</sup>see Appendix A for a detailed comparison

income, such that purely idiosyncratic income shocks move expectations about aggregates. Motivated by this evidence, this paper models households as observing variables entering their own budget constraints and treating them as signals about aggregate equilibrium objects, which they process through Bayesian updating.

There is a large literature bringing dispersed information into theoretical models. One strand of this literature focuses on explicitly linear models, or ones that are linearized as part of the solution. The seminal work is by Lucas (1972), who studies monetary non-neutrality in an island economy where agents cannot perfectly distinguish aggregate from local shocks. Relatedly, Lorenzoni (2009) studies dispersed signals and demand shocks in a business-cycle model, while Angeletos and La’O (2020), Kohlhas and Walther (2021), and related work analyze how dispersed information shapes expectations and aggregate dynamics. In linear models, Nimark (2017) as well as Huo and Takayama (2024) show that even models with higher-order belief formation can be solved under rational expectations without additional approximations.

Another strand of the theoretical literature relaxes the rational expectations assumption. In addition to generating a variety of results, this allows researchers to tackle more complex or nonlinear models. Closely related to the dispersed information literature discussed previously is work by Bordalo et al. (2018), Angeletos and La’O (2013), Angeletos et al. (2021), and Guarda (2026). There is also a large literature that deviates from rational expectations by considering subjective beliefs. Marcet and Sargent (1989) and Evans and Honkapohja (2001) implement this through adaptive learning about the underlying model process. A separate strand considers “internal rationality” based on work by Adam and Marcet (2011). Adam et al. (2016) as well as Adam et al. (2017) use this concept to implement heterogeneity in expectations. Relative to dispersed information, this approach differs in that signals are common, but agents hold subjective beliefs about the underlying process.

The paper also relates to work on sluggish expectations and excess smoothness. Sticky-information models, including Mankiw and Reis (2002), Carroll (2003), Carroll et al. (2020), and Auclert et al. (2023), generate slow-moving expectations by modeling agents that update their information sets only occasionally. A related literature, including Campbell and Deaton (1989) and Dynan (2000), studies excess smoothness and habit formation as sources of gradual adjustment. My paper shares these sluggish aggregate dynamics, but

the mechanism is different. Expectations adjust slowly because households learn gradually from dispersed signals about aggregates, not because updating is infrequent or preferences feature habit formation.

The fiscal information channel in this paper relates to evidence that expansionary policy announcements can induce pessimistic revisions in expectations. Metiu and Stockerl (2024), on whose work this paper builds, use the same BOP-HH experiment and document the increase in pessimism following information about the fiscal package. Evidence from fiscal announcements in Japan also supports an adverse information channel, particularly when unexpectedly large stimulus announcements signal weak economic conditions (Melosi et al., 2025). Related evidence comes from Coibion et al. (2022), who find that informing households about the fiscal response to the COVID-19 pandemic raises short-run unemployment expectations and lowers expected household income growth. This paper connects such adverse information effects to household consumption decisions and quantifies their implications for countercyclical transfers.

At the household level, there is significant evidence that cash receipts induce strong consumption responses (Souleles, 1999; Kueng, 2018). Selected estimates of short-run MPCs out of stimulus payments are smaller than these non-stimulus benchmarks (Orchard et al., 2025, 2026; Parker et al., 2022). During COVID-19, Goldfayn-Frank et al. (2022) also find a modest household spending response to the German *Kinderbonus*, with an overall MPC of 5.4% across its three payments. These comparisons are suggestive, since consumption measures, payment design, recipient characteristics, and estimation methods differ across studies. Appendix A.1 presents a suggestive comparison of selected estimates and discusses differences in payment size, spending horizons, and policy context. The information mechanism in this paper provides a reason why the countercyclical policy context itself can reduce household spending responses.

At the aggregate level, Taylor (2009, 2022) questions the effectiveness of stimulus payments in raising consumption. Orchard et al. (2025) find that the 2008 rebate payments had modest effects on aggregate consumption, driven by lower micro MPC estimates as well as general-equilibrium crowding out. Ramey (2026) documents limited aggregate effects of temporary cash transfers across several developed economies. Appendix A.2 illustrates these limited aggregate effects. My analysis complements this evidence by developing and quantifying an information channel through which countercyclical transfers can induce

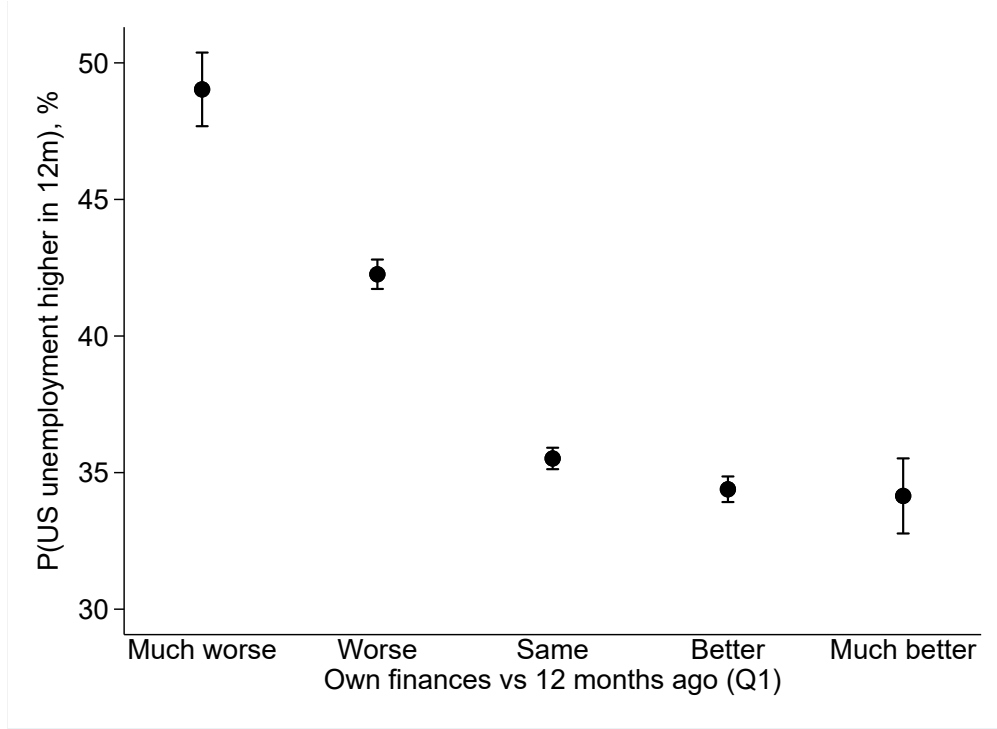
pessimistic revisions in expectations and lower aggregate consumption responses.

More broadly, the present paper relates to work on information transmission through fiscal policy, for example by Melosi et al. (2025) and Georgarakos and Kenny (2022), and to the large literature on the signaling effects of monetary policy, including Nakamura and Steinsson (2018), Melosi (2017), and Kohlhas (2022).

The main method used to solve the model is an extension of the one developed by Krusell and Smith (1998). In line with work by Marcet and Sargent (1989), this can be viewed as solving for a restricted perceptions equilibrium (RPE). Against this backdrop, the present paper can be viewed as solving for an RPE that approximates the full rational expectations equilibrium (REE). A nonparametric RPE is also used in a setting with information frictions by Guarda (2026), albeit not with the goal of approximating the REE. Also related is work by Broer et al. (2026), who apply this equilibrium concept to a model of information acquisition.

On the general topic of solving heterogeneous-agent models with dispersed information, this paper is closely related to ongoing work by Auclert et al. (2026) and Overhage (2026), who develop differing approaches to solving this class of models in sequence space. Both of these papers solve for local, linear dynamics around a full-information steady state. Relative to these, the solution method in this paper differs in that it is a global solution method resting on an assumption about expectation formation.

Lastly, this paper relates to Moll (2024), who proposes replacing rational expectations with beliefs over a more limited set of aggregate variables. One of his proposed solutions is to model expectation formation in line with survey evidence. The present paper demonstrates that an information friction can microfound this even in an approximate rational expectations equilibrium, and proposes a tractable solution method for it.



**Figure 1:** Aggregate unemployment expectations by household financial situation

### 3 Empirical evidence

#### 3.1 Experience effects and forecast efficiency

The SCE asks respondents whether their household is financially better or worse off than twelve months earlier. The response, denoted by  $Q1_{it}$ , is recorded on a five-point scale from 1, much worse, to 5, much better. Respondents also report the percent chance that the aggregate unemployment rate will be higher twelve months later, denoted by  $Q4new_{it}$ .

Figure 1 plots weighted mean unemployment expectations across the five financial-situation categories. Respondents who report that their household is financially worse off assign a substantially higher probability to an increase in aggregate unemployment. The relationship is monotonic across the financial-situation categories.

To quantify this relationship, I estimate

$$Q4new_{it} = \alpha_i + \delta_t + \beta Q1_{it} + \varepsilon_{it}, \quad (1)$$

**Table 1:** Own financial situation and expected aggregate unemployment

|                                  | P(unemployment higher) |
|----------------------------------|------------------------|
| No fixed effects                 | -4.13***<br>(0.12)     |
| Month fixed effects              | -3.89***<br>(0.12)     |
| Respondent + month fixed effects | -1.58***<br>(0.10)     |
| Weighted, respondent + month FE  | -1.60***               |
| Observations                     | 185,086                |

where  $\alpha_i$  denotes respondent fixed effects and  $\delta_t$  denotes survey-month fixed effects. Month fixed effects absorb aggregate conditions and public information shared by respondents interviewed in the same month. Respondent fixed effects absorb persistent differences in beliefs and reporting behavior. The regressions use SCE survey weights, and standard errors are clustered by respondent.

The full fixed-effects ladder, including the pooled and respondent-fixed-effects-only specifications and the corresponding unweighted estimates, is reported in Appendix B.2. The first specification includes month fixed effects. A one-category improvement in the household's financial situation is associated with an approximately 3.9 percentage point decline in the reported probability that unemployment will rise. Because this comparison is made within survey month, the estimate cannot be explained by financially worse-off households simply being observed during periods of weak aggregate conditions.

The second specification adds respondent fixed effects. The coefficient falls to approximately  $-1.6$  percentage points but remains statistically significant. The smaller within-respondent estimate is consistent with the short length of the SCE panel and the limited variation in respondents' broad financial-status categories during their time in the survey.

These results imply that households interviewed at the same time do not hold identical expectations about a purely aggregate outcome. Personal financial circumstances predict disagreement even after conditioning on common aggregate information. This is consistent with dispersed information and experience-based inference.

I obtain complementary evidence from direct measures of household income. In the

BOP-HH, declines in real household income are associated with lower expectations of aggregate growth and higher expectations of aggregate unemployment. The relationship is substantially stronger for income losses than for income gains. This is likely due to much of the data being collected during times of high inflation, in which nominal income increases do not necessarily imply real ones. Appendix B.2 reports the full BOP-HH income specification.

The SCE relationship between personal finances and aggregate beliefs does reject full information, but not necessarily rational expectations. Households may hold different forecasts because they observe different signals, while still using those signals efficiently. I therefore compare the relationship of financial conditions with respondents' forecasts and with subsequent aggregate realizations.

Define the forecast as

$$F_{it} = Q4new_{it}/100, \quad (2)$$

and the realization as

$$R_{it} = \mathbf{1}\{u_{t+12} > u_t\}, \quad (3)$$

where  $u_t$  is the US unemployment rate in the interview month. The forecast error is

$$FE_{it} = R_{it} - F_{it}. \quad (4)$$

For each outcome  $d_{it} \in \{F_{it}, R_{it}, FE_{it}\}$ , I estimate

$$d_{it} = \alpha + \beta_d Q1_{it} + \varepsilon_{it}. \quad (5)$$

The regressions use SCE survey weights and two-way clustering by survey month and respondent. Clustering by month is important because the realized outcome is common to all respondents interviewed in the same month. Respondent clustering allows forecast

**Table 2:** Forecast efficiency with respect to household finances

|                                          | Slope on own finances  |
|------------------------------------------|------------------------|
| Forecast                                 | -0.0391***<br>(0.0021) |
| Realization                              | -0.0362***<br>(0.0105) |
| Forecast error (realized minus forecast) | 0.0029<br>(0.0108)     |
| Observations                             | 180,863                |

and reporting errors to remain correlated within individuals.

A one-category improvement in financial conditions is associated with a 3.9 percentage point decline in the forecasted probability of higher unemployment. The corresponding realization slope is a 3.6 percentage point decline. The forecast-error slope is 0.3 percentage points and is not statistically distinguishable from zero. Appendix B.2 reports the corresponding estimates under alternative one-way clustering conventions.

The similarity of the forecast and realization coefficients implies that the differences in unemployment expectations across financial-situation categories broadly reflect differences in the probability of unemployment subsequently increasing. This is an efficiency result rather than evidence of full information. Households may hold different forecasts because they possess different information, while their forecast errors remain orthogonal to that information. Taken together, the evidence is consistent with rational updating under dispersed information.

### 3.2 The fiscal-package signal

I next examine whether a fiscal-policy announcement conveys information about aggregate conditions and affects household spending plans. The analysis uses a randomized information experiment conducted in the April 2020 wave of the BOP-HH (Deutsche Bundesbank, 2020). A detailed discussion of this experiment is provided by Metiu and Stockerl (2024). Respondents were assigned to one of six information treatments. The analysis in this section compares respondents receiving information about the German

**Table 3:** Policy information and expected GDP growth

|                             | Expected GDP growth |
|-----------------------------|---------------------|
| Fiscal package (arm 5)      | -1.76***<br>(0.56)  |
| Adjusted placebo prediction | -1.86               |
| Unweighted estimate         | -1.37***            |
| Controls                    | Yes                 |
| Observations                | 628                 |

Source: Research Data and Service Centre (RDSC) of the Deutsche Bundesbank, Bundesbank Online Panel – Households (BOP-HH), April 2020, own calculations.

federal fiscal package, arm 5, with respondents receiving a placebo text about European human rights, arm 1. The remaining experimental arms are excluded. Covariate balance for the fiscal-package and placebo groups is reported in Appendix B.3.

Immediately after treatment, respondents report a probability distribution over German GDP movements during the following twelve months. I convert this distribution into a mean growth forecast using the midpoints of the eight response bins. Respondents also report propensities from 0 to 100 to purchase a car, purchase another large durable good, and increase saving.<sup>2</sup>

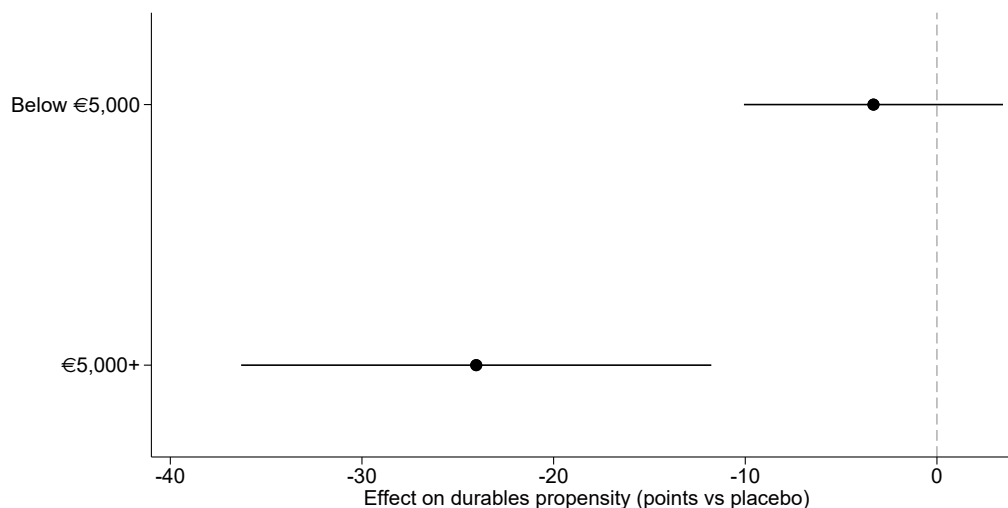
For the effect on GDP expectations, I estimate

$$GDP_i^e = \alpha + \beta f p_i + X_i' \lambda + \varepsilon_i, \quad (6)$$

where  $f p_i$  equals one for the fiscal-package treatment and zero for the placebo treatment. The vector  $X_i$  contains gender, age, household size, employment status, and pre-treatment growth expectations. The regression uses BOP-HH survey weights and heteroskedasticity-robust standard errors.

The fiscal-package announcement lowers expected GDP relative to the placebo treatment. Respondents therefore interpret the announcement not only as information about the government's policy response, but also as information that underlying aggregate condi-

<sup>2</sup>Note that this question is somewhat ambiguous, as there is no distinction between risky and safe or short- versus long-run assets.



**Figure 2:** Fiscal-package information and intended durable spending by household income

Source: Research Data and Service Centre (RDSC) of the Deutsche Bundesbank, Bundesbank Online Panel – Households (BOP-HH), April 2020, own calculations.

tions are weak. It should be noted that identifying this negative information channel is not this paper’s contribution, as this effect was already identified by Metiu and Stockerl (2024). Overall, informing households about the size of the fiscal package induces a similar, even slightly higher degree of pessimism than informing them about general economic conditions. Figure 16 in Appendix B.3 places this effect in the context of all six experimental arms. Additionally, Appendix B.5 shows that the effect on expectations is persistent, although there is significant attrition in the panel over subsequent survey waves.

I next examine whether the treatment affects intended durable spending differently across the household-income distribution. Figure 2 reports treatment effects separately for households with monthly income below EUR 5,000 and households at or above EUR 5,000.

The fiscal-package treatment has little effect on intended durable spending among households below the threshold but produces a substantially larger decline among households with income of at least EUR 5,000. The difference suggests that the same negative aggregate signal produces heterogeneous spending responses, even among groups whose beliefs move in broadly similar directions.

A natural interpretation is that higher-income households have greater scope to adjust discretionary durable purchases in response to negative news. The result is therefore

consistent with a common information update combined with heterogeneous marginal spending responses. Appendix B.4 reports the income-bracket profiles.

The belief effect also remains negative in subsequent BOP-HH waves, although the estimates become less precise as the experimental panel attrits. Figure 18 in Appendix B.5 reports the follow-up results.

Overall, the experiment supports the interpretation of fiscal policy as both an economic intervention and an information signal. Information about the fiscal package lowers expected growth and reduces intended spending among households with substantial capacity to adjust durable purchases. The responses are strongest among households for whom the announcement is relatively surprising. These results motivate the model's assumption that fiscal transfers and other components of the household budget constraint contain information about the aggregate state.

## 4 Information mechanism

This section uses a simple partial-equilibrium consumption–savings environment to isolate the information mechanism underlying the quantitative model below. Households infer aggregate productivity from their own income, which contains both an aggregate and an idiosyncratic component. The setup highlights two implications. First, cash flows affect consumption through both resources and information. Second, gradual learning about aggregate shocks can generate delayed aggregate responses.

### 4.1 Setup and signal extraction

Time is discrete,  $t = 0, 1, 2, \dots$ . There is a unit mass of identical firms and a unit mass of households. Labor supply is inelastic and normalized to one. Firms produce with linear technology

$$Y_t = Z_t, \tag{7}$$

where, denoting log variables with a lower case, aggregate productivity follows

$$z_{t+1} = \rho_z z_t + \mu_{t+1}, \quad \mu_{t+1} \sim \mathcal{N}(0, \sigma_\mu^2). \quad (8)$$

Under perfect competition and constant returns, a common wage  $W_t = Z_t$  would reveal the aggregate state exactly. Instead, I introduce wage scrambling following the idea of Boppart et al. (2024). Individual wages are dispersed deviations from aggregate productivity:

$$\int_0^1 W_{it} di = Z_t, \quad w_{it} = z_t + \xi_{it} - \frac{\sigma_\xi^2}{2}, \quad \xi_{it} \sim \mathcal{N}(0, \sigma_\xi^2). \quad (9)$$

The adjustment by  $-\sigma_\xi^2/2$  ensures that average wages remain equal to  $Z_t$ .

Households have CRRA preferences,

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \frac{C_{it}^{1-\sigma}}{1-\sigma}, \quad (10)$$

and access to a storage technology with exogenous gross return  $R^*$ . Their budget constraint is

$$C_{it} + A_{i,t+1} = R^* A_{it} + W_{it}, \quad A_{i,t+1} \geq \bar{a}, \quad \bar{a} = 0. \quad (11)$$

Households do not observe  $Z_t$  directly. They observe their own wage and understand the processes for  $Z_t$  and  $\xi_{it}$ . Defining

$$s_{it} \equiv w_{it} + \frac{\sigma_\xi^2}{2}, \quad (12)$$

the state and signal processes are

$$z_{t+1} = \rho_z z_t + \mu_{t+1}, \quad \mu_{t+1} \sim \mathcal{N}(0, \sigma_\mu^2), \quad (13)$$

$$s_{it} = z_t + \xi_{it}, \quad \xi_{it} \sim \mathcal{N}(0, \sigma_\xi^2). \quad (14)$$

Individual income therefore provides a noisy signal about aggregate productivity.

The system is linear and Gaussian, so households apply the Kalman filter. Let  $\hat{z}_{i,t|t-1}$

denote household  $i$ 's prior estimate of  $z_t$ . After observing  $s_{it}$ ,

$$\hat{z}_{i,t|t} = (1 - k_t)\hat{z}_{i,t|t-1} + k_t s_{it}, \quad (15)$$

$$k_t = \frac{P_{t|t-1}}{P_{t|t-1} + \sigma_\xi^2}, \quad (16)$$

where  $P_{t|t-1}$  is the prior variance. The posterior variance follows the standard Kalman recursion and converges to a common stationary value. Because  $k_t \in (0, 1)$ , households only partially incorporate an income realization into their estimate of the aggregate state. Beliefs therefore adjust gradually to aggregate shocks.

Define cash on hand as

$$M_{it} = R^* A_{it} + W_{it}. \quad (17)$$

Once the posterior variance has converged, the household problem can be written using  $(M_{it}, \hat{z}_{i,t|t})$  as individual states:

$$V(M_{it}, \hat{z}_{i,t|t}) = \max_{A_{i,t+1} \geq \bar{a}} \left\{ u(C_{it}) + \beta \mathbb{E}_{it} \left[ V(M_{i,t+1}, \hat{z}_{i,t+1|t+1}) \right] \right\}, \quad (18)$$

where

$$C_{it} = M_{it} - A_{i,t+1}, \quad (19)$$

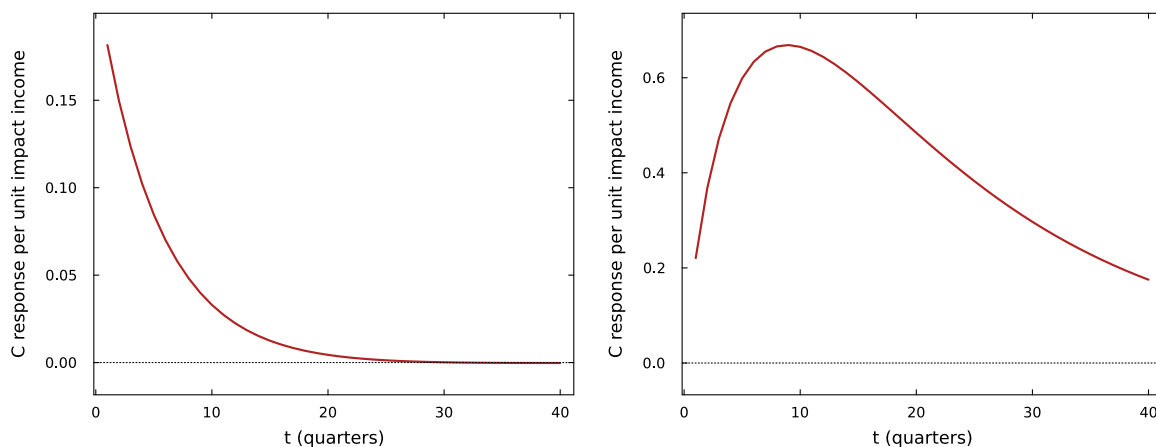
$$M_{i,t+1} = R^* A_{i,t+1} + W_{i,t+1}. \quad (20)$$

An observed income change therefore affects consumption through current cash on hand as well as through beliefs about future income.

## 4.2 Consumption implications

The information friction has two closely related implications for consumption dynamics. Aggregate shocks are incorporated into household beliefs only gradually, generating delayed aggregate adjustment. At the household level, the response to an observed income change depends not only on the additional resources it provides, but also on what it reveals about the aggregate state.

Figure 3 illustrates the first mechanism.



**Figure 3:** Consumption responses to one-standard-deviation shocks. Notes: percentage deviations from steady state.

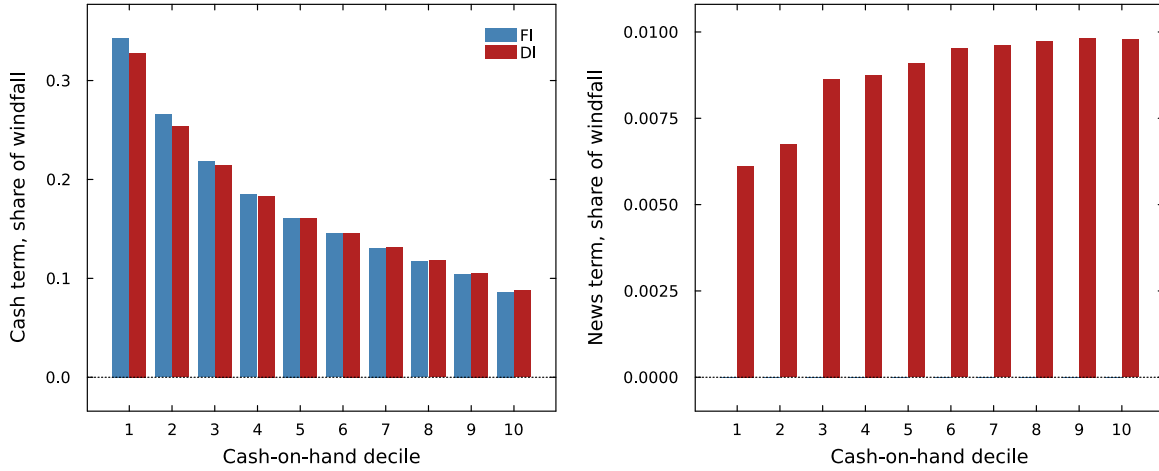
The left panel shows the response of aggregate consumption to a one-standard-deviation aggregate productivity shock. On impact, consumption rises by about half of its eventual peak, with the maximum effect occurring only several quarters later. The right panel shows the consumption response of an individual household to a one-standard-deviation idiosyncratic wage shock, evaluated at median cash on hand. This response is instead front-loaded.

The contrast reflects signal extraction. Aggregate shocks must first pass through dispersed income signals and Bayesian updating before they are fully incorporated into household beliefs, so their effect on aggregate consumption is spread over time. An idiosyncratic income innovation instead changes household resources immediately. This generates a front-loaded individual response alongside a delayed aggregate response.

The same mechanism implies that MPCs depend on the informational content of an income change. It is useful to distinguish between a pure cash-flow response, which changes current resources while holding beliefs fixed, and an informational component operating through the household's posterior belief about aggregate productivity.

Figure 4 illustrates this decomposition.

As in standard incomplete-markets models, the pure cash-flow response is largest for households with little cash on hand and declines along the distribution. Under dispersed



**Figure 4:** Quarterly responses to an idiosyncratic income shock

information, however, the same income realization also changes expectations. A positive wage innovation is partly interpreted as evidence that aggregate productivity, and therefore future income, is higher. The total consumption response can therefore differ from the pure MPC that holds beliefs fixed.

In this basic model, these effects are relatively minor overall. The information effect is less pronounced at the bottom of the distribution in absolute terms, and especially relative to the cash component. Constraints play the main role here, with households closer to the borrowing constraint less sensitive to expectations about future income. It is only at the top of the distribution that the information channel is about 10% as big as the cash component. While the size is small in this basic framework, the pattern of information mattering more at the top of the distribution holds true in the more involved model from Section 5 onward.

The environment therefore isolates the two mechanisms used in the fiscal-policy model below. Dispersed information delays aggregate adjustment because households learn about aggregate conditions over time, while the response to a cash flow depends on what that cash flow reveals about future economic conditions. In the fiscal-policy model, transfers themselves play this dual role as both resources and signals.

## 5 Fiscal-policy model

This section introduces fiscal transfers into the heterogeneous-household model with dispersed information and studies their quantitative effects. As in the basic model, households have fixed labor supply and face persistent idiosyncratic income risk. Transfers provide households with resources, but also convey information about aggregate conditions. Aggregate output responds to household consumption through a reduced-form demand externality, following, for example, Boppart et al. (2018) and Krueger et al. (2016).

Households observe the variables entering their own budget constraints, but cannot separate aggregate output from the idiosyncratic component of their labor income. They also observe aggregate transfers, but cannot determine whether a high transfer reflects a discretionary fiscal intervention or the systematic countercyclical response of fiscal policy. Households therefore use labor income and transfers to infer aggregate output, their own idiosyncratic efficiency, and the discretionary component of fiscal policy.

### 5.1 Environment

There is a unit mass of households indexed by  $i \in [0, 1]$ . Households save in an internationally traded asset with exogenous gross return

$$R^* = 1 + r^*, \quad \beta R^* < 1. \quad (21)$$

Aggregate productivity follows the log AR(1) process

$$z_t = \rho_z z_{t-1} + \mu_t, \quad \mu_t \sim \mathcal{N}(0, \sigma_\mu^2), \quad |\rho_z| < 1, \quad (22)$$

where  $z_t = \log Z_t$ .

Households differ in persistent idiosyncratic efficiency. Let

$$e_{i,t+1} = \rho_e e_{it} + \zeta_{i,t+1}, \quad \zeta_{i,t+1} \sim \mathcal{N}(\mu_\zeta, \sigma_\zeta^2), \quad |\rho_e| < 1, \quad (23)$$

where  $e_{it} = \log E_{it}$ . The stationary variance of log efficiency is

$$\sigma_e^2 = \frac{\sigma_\zeta^2}{1 - \rho_e^2}. \quad (24)$$

I normalize mean efficiency in levels to one:

$$\mathbb{E}[E_{it}] = 1. \quad (25)$$

Since  $e_{it}$  is normally distributed, this requires

$$\mathbb{E}[e_{it}] = -\frac{\sigma_e^2}{2}. \quad (26)$$

The corresponding drift in the innovation process is

$$\mu_\zeta = -(1 - \rho_e) \frac{\sigma_e^2}{2} = -\frac{\sigma_\zeta^2}{2(1 + \rho_e)}. \quad (27)$$

With a continuum of households, the normalization implies

$$\int_0^1 E_{it} di = 1. \quad (28)$$

Household  $i$  has preferences

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(C_{it}), \quad u(C) = \frac{C^{1-\sigma}}{1-\sigma}. \quad (29)$$

Labor supply is fixed. Household  $i$  receives individual labor income  $W_{it}$ , an aggregate transfer  $T_t$ , and pays a lump-sum tax  $\tau_t$ . Its budget constraint is

$$C_{it} + A_{i,t+1} = R^* A_{it} + W_{it} + T_t - \tau_t, \quad (30)$$

subject to

$$A_{i,t+1} \geq 0. \quad (31)$$

Cash on hand is therefore

$$M_{it} = R^* A_{it} + W_{it} + T_t - \tau_t. \quad (32)$$

Aggregate consumption is

$$C_t = \int_0^1 C_{it} di, \quad c_t = \log C_t. \quad (33)$$

Aggregate output is determined by productivity and the aggregate consumption externality:

$$Y_t = Z_t C_t^\omega, \quad \omega \geq 0. \quad (34)$$

Define  $y_t = \log Y_t$ . The production relation in logs is

$$y_t = z_t + \omega c_t. \quad (35)$$

Aggregate labor income equals output,

$$W_t = Y_t, \quad (36)$$

and individual labor income is

$$W_{it} = Y_t E_{it}. \quad (37)$$

Consequently, observed individual log labor income satisfies

$$w_{it} = y_t + e_{it}, \quad (38)$$

where  $w_{it} = \log W_{it}$ . When  $\omega = 0$ , aggregate output is determined entirely by productivity. When  $\omega > 0$ , higher household consumption raises output and labor income, which feeds back into household consumption.

## 5.2 Fiscal policy

Fix an output reference level  $Y^{\text{ref}}$  and define

$$y^{\text{ref}} = \log Y^{\text{ref}}, \quad \tilde{y}_t = \log \left( \frac{Y_t}{Y^{\text{ref}}} \right) = y_t - y^{\text{ref}}. \quad (39)$$

Normalized aggregate transfers are

$$\tilde{T}_t = \frac{T_t}{Y^{\text{ref}}}. \quad (40)$$

Transfers respond countercyclically to output and contain a persistent discretionary component:

$$\tilde{T}_t = -\psi_y \tilde{y}_t + \nu_t, \quad \nu_t = \rho_\nu \nu_{t-1} + \varepsilon_t^\nu, \quad \varepsilon_t^\nu \sim \mathcal{N}(0, \sigma_\nu^2). \quad (41)$$

Equivalently,

$$T_t = Y^{\text{ref}} \left[ -\psi_y (y_t - y^{\text{ref}}) + \nu_t \right]. \quad (42)$$

A high transfer may therefore reflect weak aggregate output or a positive discretionary fiscal state:

$$\tilde{T}_t \uparrow \implies y_t \downarrow \quad \text{or} \quad \nu_t \uparrow. \quad (43)$$

Government debt evolves according to

$$B_{t+1}^g = R^* B_t^g + T_t - \tau_t. \quad (44)$$

Define government debt relative to its reference level as

$$D_t = B_t^g - \bar{B}. \quad (45)$$

Taxes stabilize government debt according to

$$\tau_t = \bar{\tau} + \kappa D_t. \quad (46)$$

At the reference point,  $\tilde{y}_t = 0$ ,  $\nu_t = 0$ , and  $B_t^g = \bar{B}$ , so transfers are zero and

$$\bar{\tau} = (R^* - 1)\bar{B}. \quad (47)$$

Combining the government budget constraint with the fiscal rules gives

$$D_{t+1} = (R^* - \kappa)D_t + Y^{\text{ref}} \left[ -\psi_y (y_t - y^{\text{ref}}) + \nu_t \right]. \quad (48)$$

Because taxes are observed, households infer the current debt deviation exactly:

$$D_t = \frac{\tau_t - \bar{\tau}}{\kappa}. \quad (49)$$

Government debt is therefore an observed aggregate state, rather than an additional hidden signal.

### 5.3 Information structure and timing

Households observe all variables entering their current budget constraints:

$$\mathcal{I}_{it} \supseteq \{A_{it}, W_{it}, T_t, \tau_t\}. \quad (50)$$

They do not directly observe aggregate output  $y_t$ , idiosyncratic efficiency  $e_{it}$ , or the discretionary fiscal state  $\nu_t$ . Aggregate productivity  $z_t$  and its innovation  $\mu_t$  are also unobserved. The two signals about hidden current conditions are

$$w_{it} = y_t + e_{it}, \quad (51)$$

$$\tilde{T}_t = -\psi_y y_t + \psi_y y^{\text{ref}} + \nu_t. \quad (52)$$

Labor income confounds aggregate output with idiosyncratic efficiency. Transfers confound discretionary fiscal policy with the systematic countercyclical response to aggregate output. Taxes reveal  $D_t$  exactly through (49).

The timing within period  $t$  is as follows. Aggregate productivity and the fiscal innovation are realized. Given predetermined government debt and the cross-sectional distribution of household states, aggregate consumption, output, transfers, and labor income are determined jointly. Household  $i$  observes  $(W_{it}, T_t, \tau_t)$ , updates its beliefs, and chooses consumption and next-period assets.

Absent an approximation, the aggregate state includes productivity, government debt, the fiscal state, and the cross-sectional distribution of household assets, efficiencies, and beliefs. Household beliefs affect saving, saving determines aggregate consumption, aggregate consumption affects output, and output changes the signals households use to form beliefs.

The exact forecasting problem is therefore infinite dimensional.

## 5.4 Perceived law of motion and Bayesian updating

I approximate aggregate log output directly with the perceived law of motion

$$y_t = b_0 + b_y y_{t-1} + b_B D_t + b_v v_t + b_\mu \mu_t. \quad (53)$$

The productivity innovation enters the perceived law directly, rather than productivity being an additional perceived aggregate state. The law in (53) is a restricted approximation to the simulated equilibrium, not an algebraic consequence of (34).

Because the current transfer and tax payment determine next-period debt exactly,  $D_{t+1}$  is known at the end of period  $t$ . Under the perceived law, output evolves according to

$$y_{t+1} = b_0 + b_y y_t + b_B D_{t+1} + b_v \rho_v v_t + b_\mu \mu_{t+1} + b_v \varepsilon_{t+1}^v. \quad (54)$$

For filtering, it is convenient to collect the hidden states in

$$q_{it} = (y_t, e_{it}, v_t)'. \quad (55)$$

Define the centered idiosyncratic innovation

$$\varepsilon_{i,t+1}^e = \zeta_{i,t+1} - \mu_\zeta, \quad \varepsilon_{i,t+1}^e \sim \mathcal{N}(0, \sigma_\zeta^2). \quad (56)$$

The perceived transition can be written as

$$q_{i,t+1} = d_q + A_q q_{it} + D_q D_{t+1} + G_q \varepsilon_{i,t+1}, \quad (57)$$

where

$$d_q = \begin{pmatrix} b_0 \\ \mu_\zeta \\ 0 \end{pmatrix}, \quad D_q = \begin{pmatrix} b_B \\ 0 \\ 0 \end{pmatrix}, \quad (58)$$

$$A_q = \begin{pmatrix} b_y & 0 & b_v \rho_v \\ 0 & \rho_e & 0 \\ 0 & 0 & \rho_v \end{pmatrix}, \quad (59)$$

and

$$\boldsymbol{\varepsilon}_{i,t+1} = \begin{pmatrix} \mu_{t+1} \\ \varepsilon_{t+1}^v \\ \varepsilon_{i,t+1}^e \end{pmatrix}, \quad G_q = \begin{pmatrix} b_\mu & b_v & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}. \quad (60)$$

The covariance matrix of the primitive innovations is

$$\Sigma_\varepsilon = \begin{pmatrix} \sigma_\mu^2 & 0 & 0 \\ 0 & \sigma_v^2 & 0 \\ 0 & 0 & \sigma_\zeta^2 \end{pmatrix}, \quad Q_q = G_q \Sigma_\varepsilon G_q'. \quad (61)$$

Define the observation vector as

$$s_{it} = (w_{it}, \tilde{T}_t)'. \quad (62)$$

The two exact observation equations are

$$s_{it} = h + Hq_{it}, \quad h = \begin{pmatrix} 0 \\ \psi_y y^{\text{ref}} \end{pmatrix}, \quad H = \begin{pmatrix} 1 & 1 & 0 \\ -\psi_y & 0 & 1 \end{pmatrix}. \quad (63)$$

Let

$$m_{i,t|t-1}^q = \mathbb{E}_{it-1}[q_{it}] \quad (64)$$

denote household  $i$ 's prior mean and let  $P_{t|t-1}$  denote the corresponding prior covariance matrix. Since  $D_t$  is observed, the prediction step is

$$m_{i,t|t-1}^q = d_q + A_q m_{i,t-1|t-1}^q + D_q D_t, \quad (65)$$

$$P_{t|t-1} = A_q P_{t-1|t-1} A_q' + Q_q. \quad (66)$$

After observing  $s_{it}$ , the Kalman gain is

$$K_t = P_{t|t-1} H' \left( H P_{t|t-1} H' \right)^{-1}. \quad (67)$$

The posterior mean and covariance are

$$m_{i,t|t}^q = m_{i,t|t-1}^q + K_t \left( s_{it} - h - H m_{i,t|t-1}^q \right), \quad (68)$$

$$P_{t|t} = (I - K_t H) P_{t|t-1}. \quad (69)$$

The covariance matrices are common across households. For a given perceived law of motion, they converge to stationary matrices and do not enter the individual household state. The two exact observations for income leave only one independent posterior mean. I use

$$\hat{y}_{i,t|t} = \mathbb{E}_{it}[y_t] \quad (70)$$

as the reduced belief state. The posterior means of individual log efficiency and the fiscal state are implied by the current signals:

$$\hat{e}_{i,t|t} = w_{it} - \hat{y}_{i,t|t}, \quad (71)$$

$$\hat{v}_{i,t|t} = \tilde{T}_t + \psi_y(\hat{y}_{i,t|t} - y^{\text{ref}}). \quad (72)$$

One-period-ahead expectations satisfy

$$\mathbb{E}_{it}[v_{t+1}] = \rho_v \hat{v}_{i,t|t}, \quad (73)$$

$$D_{t+1} = (R^* - \kappa) D_t + T_t, \quad (74)$$

$$\mathbb{E}_{it}[e_{i,t+1}] = \mu_\zeta + \rho_e \hat{e}_{i,t|t}, \quad (75)$$

$$\mathbb{E}_{it}[y_{t+1}] = b_0 + b_y \hat{y}_{i,t|t} + b_B D_{t+1} + b_v \rho_v \hat{v}_{i,t|t}. \quad (76)$$

The deterministic transition for  $(y_t, D_t, v_t)$  implied by the perceived law and fiscal rules has transition matrix

$$\mathcal{A} = \begin{pmatrix} b_y - b_B \psi_y Y^{\text{ref}} & b_B (R^* - \kappa) & b_B Y^{\text{ref}} + b_v \rho_v \\ -\psi_y Y^{\text{ref}} & R^* - \kappa & Y^{\text{ref}} \\ 0 & 0 & \rho_v \end{pmatrix}. \quad (77)$$

The converged perceived system must satisfy

$$\rho(\mathcal{A}) < 1, \quad (78)$$

where  $\rho(\mathcal{A})$  denotes the spectral radius. This joint condition replaces a stability check based only on the partial debt coefficient  $R^* - \kappa$ .

## 5.5 Household problem and equilibrium

A convenient post-signal household state is

$$x_{it} = (A_{it}, W_{it}, T_t, \tau_t, \hat{y}_{i,t|t}). \quad (79)$$

The remaining posterior means are recovered from current observations using (71) and (72). The household problem can be written recursively as

$$V(x_{it}) = \max_{A_{i,t+1} \geq 0} \{u(C_{it}) + \beta \mathbb{E}_{it}[V(x_{i,t+1})]\}, \quad (80)$$

where current consumption is

$$C_{it} = R^* A_{it} + W_{it} + T_t - \tau_t - A_{i,t+1}. \quad (81)$$

Future household states are generated by the perceived output law, the fiscal rules, the idiosyncratic efficiency process, and the Kalman update.

For given productivity  $Z_t$ , fiscal state  $\nu_t$ , debt  $D_t$ , and the household distribution, aggregate consumption determines output through (34). Output in turn determines individual labor income and the systematic component of transfers. Current consumption, output, transfers, and posterior beliefs must therefore be determined jointly within the period. For a candidate value of  $C_t$ , implied output determines the labor-income and transfer signals, households update their beliefs and choose consumption, and the candidate must satisfy

$$C_t = \int_0^1 C(A_{it}, W_{it}(C_t), T_t(C_t), \tau_t, \hat{y}_{i,t|t}(C_t)) di. \quad (82)$$

Once this fixed point is found, output, transfers, household choices, and next-period government debt are determined.

## 5.6 Solution method

The model is solved as a Krusell–Smith fixed point in the perceived output-law coefficients  $(b_0, b_y, b_B, b_v, b_\mu)$ .

First, I guess the coefficients in (53). Given these coefficients, I construct the perceived state-space system and solve for the stationary Kalman covariance matrices and gain. I then solve the household consumption-saving problem under the perceived aggregate law.

Next, I simulate aggregate productivity, fiscal innovations, idiosyncratic efficiency, government debt, and a large panel of households. In each period, I solve the within-period aggregate-consumption fixed point in (82), then update household assets, efficiencies, and posterior beliefs using the household policy functions and the Kalman recursion.

Using the simulated time series, I estimate

$$y_t = b_0 + b_y y_{t-1} + b_B D_t + b_v \nu_t + b_\mu \mu_t + u_t^y. \quad (83)$$

The estimated coefficients update the perceived law of motion. This outer loop is repeated until the perceived coefficients coincide with those generated by the simulated equilibrium. I verify joint fiscal-output stability using (78). Approximation accuracy is evaluated using the fit of the aggregate output law, iterated forecasts, and household-level signal-extraction checks.

I next study how dispersed information changes the effects of fiscal transfers. I first describe the calibration and the behavior of household beliefs in the simulated equilibrium. I then compare the response to a one-standard-deviation discretionary fiscal innovation under full and dispersed information.

## 5.7 Calibration

The model is calibrated at a quarterly frequency. Table 4 reports the primitive parameters used for the quantitative results.

**Table 4:** Calibration, fiscal-policy model

| Parameter                                         | Value  | Description                     | Source                                      |
|---------------------------------------------------|--------|---------------------------------|---------------------------------------------|
| <i>Households and idiosyncratic income</i>        |        |                                 |                                             |
| $\beta$                                           | 0.982  | Quarterly discount factor       | McKay and Reis (2021)                       |
| $r^*$                                             | 0.0074 | Quarterly net interest rate     | McKay and Reis (2021)                       |
| $\sigma$                                          | 1.5    | CRRA coefficient                | Attanasio and Weber (1995)                  |
| $\rho_e$                                          | 0.863  | Idiosyncratic persistence       | Guvenen (2009); Guarda (2026)               |
| $\sigma_\zeta$                                    | 0.162  | Idiosyncratic innovation s.d.   | Guvenen (2009); Guarda (2026)               |
| <i>Aggregate process and demand amplification</i> |        |                                 |                                             |
| $\rho_z$                                          | 0.95   | Aggregate persistence           | Boppart et al. (2018)                       |
| $\sigma_\mu$                                      | 0.007  | Productivity innovation s.d.    | Boppart et al. (2018)                       |
| $\omega$                                          | 0.30   | Consumption externality         | Krueger et al. (2016)                       |
| <i>Fiscal policy</i>                              |        |                                 |                                             |
| $\psi_y$                                          | 0.187  | Countercyclicality coefficient  | Reicher (2011)                              |
| $\kappa$                                          | 0.05   | Tax response to government debt | Kindberg-Hanlon (2024)                      |
| $\bar{B}$                                         | 2.0    | Reference government debt       | Normalization                               |
| $\rho_v$                                          | 0.90   | Fiscal persistence              | McKay and Reis (2021)                       |
| $\sigma_v$                                        | 0.0036 | Fiscal innovation s.d.          | McKay and Reis (2021); Leeper et al. (2010) |

The discount factor and quarterly net interest rate are set to  $\beta = 0.982$  and  $r^* = 0.0074$ . The resulting inequality  $\beta(1 + r^*) < 1$  ensures that the incomplete-markets economy does not generate unbounded asset accumulation. The CRRA coefficient is  $\sigma = 1.5$ , and households cannot borrow.

The idiosyncratic-income process uses  $\rho_e = 0.863$  and  $\sigma_\zeta = 0.162$ , based on the quarterly mapping of the earnings process estimated by Guvenen (2009) and compared with Guarda (2026). Aggregate productivity has persistence  $\rho_z = 0.95$  and innovation standard deviation  $\sigma_\mu = 0.007$ . The demand-amplification coefficient is  $\omega = 0.30$ .

Fiscal policy has countercyclicality coefficient  $\psi_y = 0.187$ . The discretionary fiscal state has persistence  $\rho_v = 0.90$  and innovation standard deviation  $\sigma_v = 0.0036$ . Taxes respond to

debt with  $\kappa = 0.05$ , and reference debt is normalized to  $\bar{B} = 2.0$ .

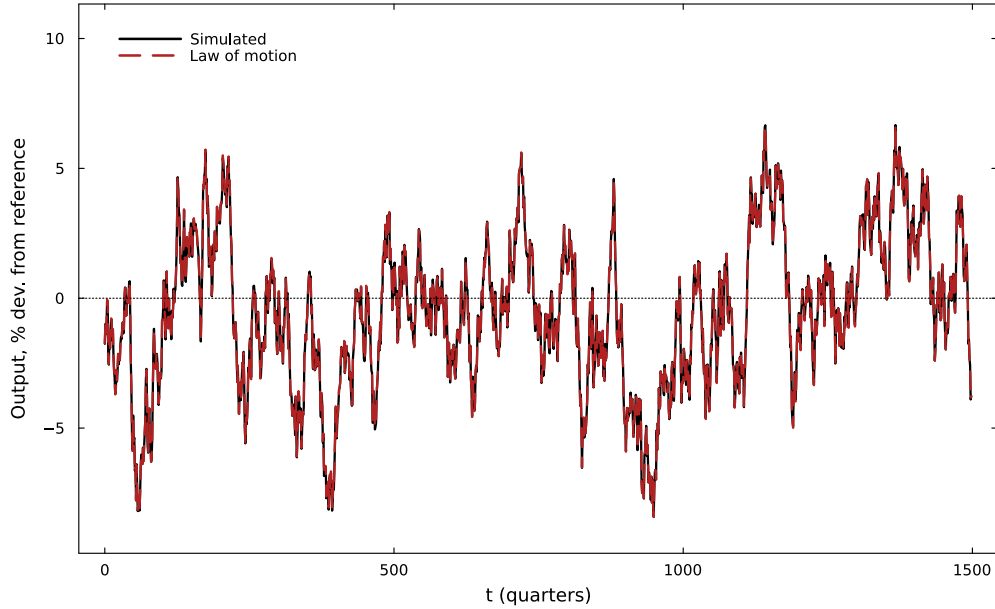
## 5.8 Aggregate approximation and household beliefs

Solving the model yields the perceived output law reported in Table 5. The aggregate approximation is very accurate, with  $R^2 = 0.9999897$ .

**Table 5:** Estimated output law of motion

|             | $b_0$                  | $b_y$    | $b_B$                     | $b_v$      | $b_\mu$ | $R^2$     |
|-------------|------------------------|----------|---------------------------|------------|---------|-----------|
| Coefficient | $1.385 \times 10^{-5}$ | 0.960833 | $-5.24010 \times 10^{-4}$ | 0.00853032 | 1.10480 | 0.9999897 |

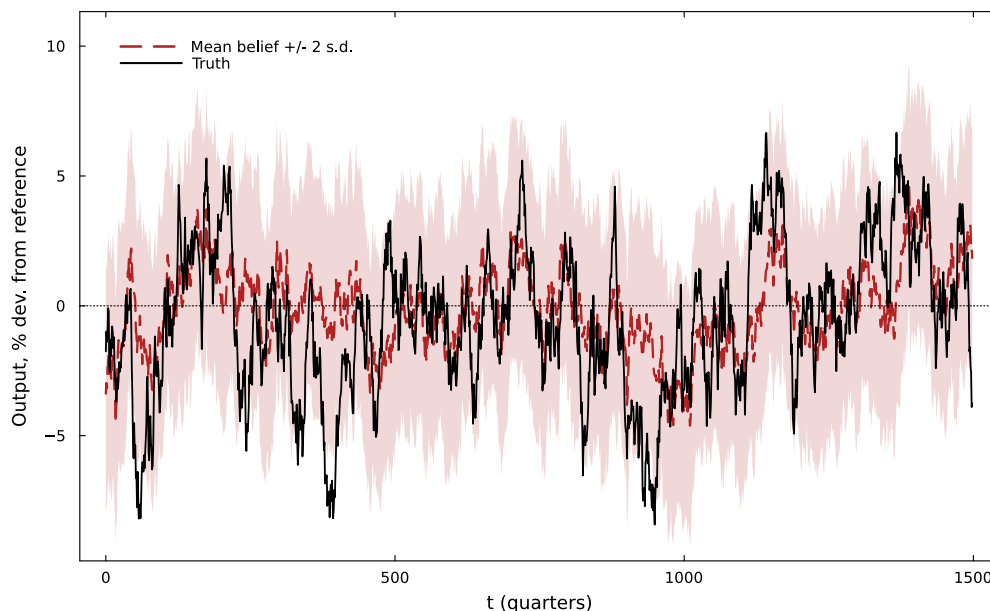
Figure 5 compares simulated output with the path implied by the converged perceived law. The two series are nearly indistinguishable. The low-dimensional law therefore captures the aggregate dynamics generated by the full distribution of household states and beliefs.



**Figure 5:** Simulated aggregate output and the perceived law of motion

This high aggregate fit does not imply that households know the current aggregate state. Households observe the signals generated by the economy, not realized aggregate output itself. Figure 6 compares true output with the cross-sectional distribution of household output beliefs. Individual posteriors remain dispersed because labor income combines

aggregate output with idiosyncratic efficiency, while transfers combine the countercyclical policy response with the discretionary fiscal state.



**Figure 6:** Aggregate output and the distribution of household beliefs

### 5.8.1 Ex-post signal-extraction checks

The perceived law of motion approximates aggregate output extremely closely, but it remains possible that the Kalman filter imposes an overly restrictive mapping from household observations to output beliefs. I assess this possibility using ex-post machine-learning counterfactuals. These exercises hold the simulated equilibrium fixed and replace the model’s filtering rule with alternative forecasting methods. They do not resolve the household problem or recompute the equilibrium under the alternative beliefs.

The alternatives receive the same current observables as households in the model:

$$o_{it} = (w_{it}, \tilde{T}_t, D_t)'. \quad (84)$$

I consider a ridge regression and a  $k$ -nearest-neighbor estimator using twelve quarters of observations, as well as a gated recurrent unit that summarizes the history recursively. Each method is trained to forecast current aggregate output  $y_t$ . I simulate 1,000 periods for 100 households and repeat the exercise 10 times.

Table 6 reports out-of-sample forecast performance. To compare with the implementation, the baseline is listed as storing the three posterior means associated with  $(y_t, e_{it}, v_t)$ . The final column reports the share of evaluated household-period forecasts for which the alternative produces a larger forecast error than the model belief.

**Table 6:** Ex-post recovery of aggregate output from household observations

| Belief rule                  | Stored states<br>per household | $R^2$ | Larger error than<br>model belief |
|------------------------------|--------------------------------|-------|-----------------------------------|
| Model baseline               | 3                              | 0.281 | N/A                               |
| GRU ( $H = 16$ )             | 16                             | 0.232 | 80%                               |
| Ridge ( $L = 12$ )           | 33                             | 0.237 | 80%                               |
| $k$ NN ( $L = 12, k = 160$ ) | 33                             | 0.194 | 88%                               |

None of the alternative forecasting methods improves on the model belief. The baseline filter achieves an  $R^2$  of 0.281, compared with 0.237 for Ridge, 0.232 for the GRU, and 0.194 for  $k$ NN. The alternatives also produce larger household-level forecast errors in at least 80% of the evaluated observations.

This result is consistent with the near-exact fit of the perceived aggregate law. *If* the perceived law of motion were exact, the linear-Gaussian structure would make the Kalman posterior mean the minimum-mean-squared-error forecast conditional on household observations. The ex-post exercise shows that the small approximation error in the perceived law is not large enough to allow different signal extraction frameworks to confer better information.

These alternatives are also infeasible as equilibrium belief rules in the current solution method. Even the smallest selected GRU carries a sixteen-dimensional recursive state. Under the discretization used to solve the household problem, incorporating this state would expand the computational burden by more than a factor of  $10^8$ . Appendix C provides the construction and details for the three counterfactual forecasting rules.

## 5.9 Validation against the fiscal-policy information experiment

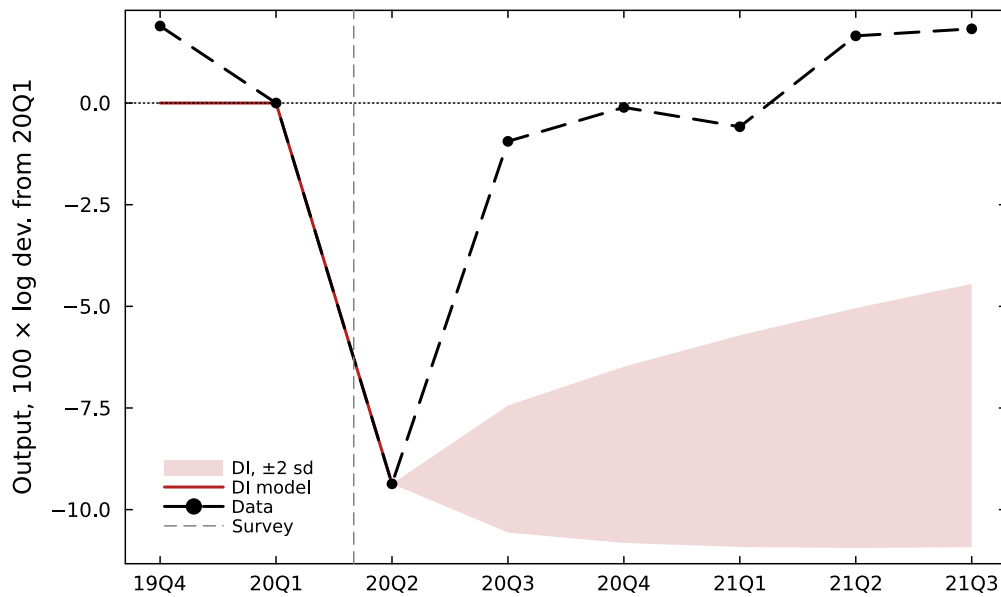
I next map the fiscal-policy information experiment into the quantitative model. For this validation exercise, I set  $Y^{\text{ref}}$  equal to Q1 2020 output. The treatment describes a

EUR 750 billion federal fiscal package, of which EUR 600 billion is support for firms (Deutsche Bundesbank, 2020). I map the remaining EUR 150 billion to the model's cumulative household-transfer component over four quarters, from Q2 2020 through Q1 2021. Expressed relative to Q1 2020 output, this model counterpart is

$$s = \sum_{j=0}^3 \tilde{T}_{t+j} = \frac{150}{884} = 0.1696. \quad (85)$$

Households observe the current transfer but are not given a quarterly allocation of the remaining package. They must therefore infer how much of the announcement reflects previously unobserved current economic conditions and how much reflects planned discretionary fiscal interventions over the following three quarters.

The exercise starts the economy at the Q2 2020 state shown in Figure 7. The vertical line denotes the timing of the survey, while the shaded region reports the conditional dispersed-information distribution following the announcement. The realized recovery was substantially faster than the model's conditional path; only a large upward shock after Q2 2020 could generate such a rapid recovery. This comparison is purely illustrative, since the exercise is used only to evaluate the information framework at the time of the decline.



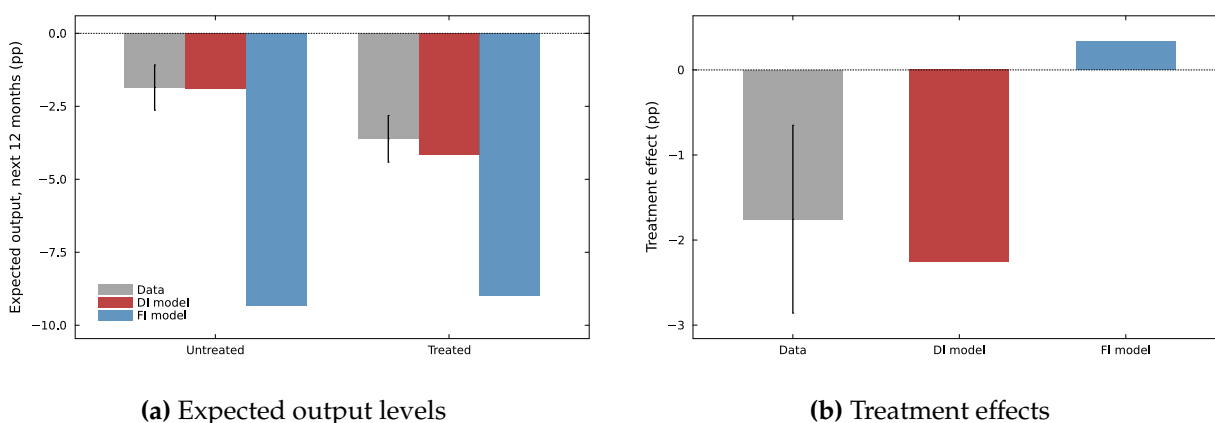
**Figure 7:** Output around the fiscal-policy information experiment

The government is assumed to know the current economic state and its own fiscal plan, but not future productivity innovations. Consequently, the announcement contains no additional information about future productivity shocks. It provides one signal about two conceptually distinct objects: the current hidden state and the sequence of planned fiscal innovations.

Households allocate the announcement surprise between these explanations according to their prior uncertainty. In the baseline calibration, approximately 1.4% of the surprise is attributed to worse current conditions and 98.6% to planned fiscal interventions. The inferred plan is front-loaded because an innovation in the first future quarter affects transfers in all three remaining quarters, whereas a later innovation affects fewer quarters. Appendix D provides the exact signal-extraction problem.

Although the current-state component is small as a share of the package surprise, it is central for GDP expectations. A downward revision in current output persists throughout the four-quarter forecast window. By contrast, planned fiscal innovations have only a small effect on expected output under the estimated perceived law of motion. They nevertheless matter for household consumption because they imply additional future transfer income.

Figure 8 compares GDP expectations in the data and model. The left panel reports expected output levels for untreated and treated households. The right panel reports the treatment effect.



**Figure 8:** GDP expectations in the experiment and model

Source: Research Data and Service Centre (RDSC) of the Deutsche Bundesbank, Bundesbank Online Panel – Households (BOP-HH), April 2020, own calculations. Model calculations by author.

The dispersed-information model produces an untreated expectation of approximately  $-1.9$  percentage points, close to the empirical value of  $-1.86$  percentage points. Following an output decline as large as the one experienced by the German economy in Q2 2020, expectations in the model adjust roughly to the same degree as those in the data.

The information treatment, or policy announcement, lowers expected output by approximately 2.3 percentage points in the model, compared with 1.76 percentage points in the data. Under full information, households already know the current state and interpret the announcement only as news about the fiscal plan. The resulting treatment effect is positive, at approximately 0.34 percentage points. The negative empirical response therefore requires households to learn something about current aggregate conditions from the announcement.

Table 7 compares the experimental spending results with the model. These outcomes are fundamentally different objects. The empirical estimates concern changes in the probability of reporting an intended purchase, whereas the model produces changes in consumption. Their magnitudes are therefore not comparable. I use the table only to evaluate whether the model reproduces the signs of the aggregate response and the corresponding heterogeneity gradients. Intuitively, the two slope rows measure whether the consumption adjustment is stronger across households that had higher incomes or more optimistic priors.

**Table 7:** Spending responses in the experiment and model

|                               | RCT               | Model                          |                           |
|-------------------------------|-------------------|--------------------------------|---------------------------|
|                               | Propensity (pp)   | Dispersed info Consumption (%) | Full info Consumption (%) |
| Aggregate spending            | $-6.15$<br>(3.09) | $-0.31 \checkmark$             | $1.43 \times$             |
| Slope across income brackets  | $-2.27$<br>(2.22) | $-0.53 \checkmark$             | $-0.40 \checkmark$        |
| Slope across prior categories | $-4.53$<br>(2.97) | $-0.60 \checkmark$             | –                         |

*Notes:* RCT entries are percentage-point effects on stated purchase propensity. Model entries are percent changes in consumption. For the slope rows, effects are measured per one-category increase. The magnitudes are not comparable, checkmarks and crosses compare signs only. Standard errors for the RCT estimates are reported in parentheses. Dashes indicate that the comparison is not defined.

Source: Research Data and Service Centre (RDSC) of the Deutsche Bundesbank, Bundesbank Online Panel – Households (BOP-HH), April 2020, own calculations. Model calculations by author.

Dispersed information reproduces the negative signs of the aggregate spending response, the income gradient, and the prior-belief gradient. Full information produces the wrong sign for the aggregate response. The negative income gradient is present under both information structures and therefore does not by itself identify the information channel. By contrast, heterogeneity by prior beliefs is inherently linked to dispersed information.

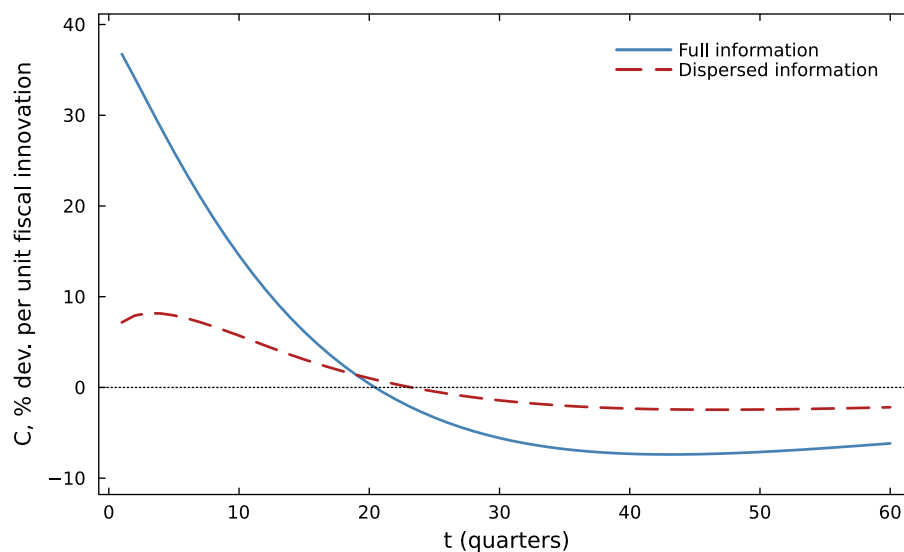
Taken together, the exercise provides two forms of validation. For GDP expectations, the empirical and model outcomes are directly comparable, and the dispersed-information model approximately reproduces both the control-group level and the negative treatment effect. For spending, the model reproduces the relevant signs but is not evaluated against the empirical magnitudes. The next subsection returns to the model's aggregate consumption response to a discretionary fiscal intervention.

## 5.10 Fiscal-policy effects

### 5.10.1 Aggregate consumption response

I next use the model to study a one-standard-deviation positive innovation to the discretionary fiscal state,  $\varepsilon_t^v = \sigma_v$ . Because  $v_t$  is persistent, the innovation raises both current transfers and expected future transfers. The shock is evaluated from the same resting distribution under full and dispersed information.

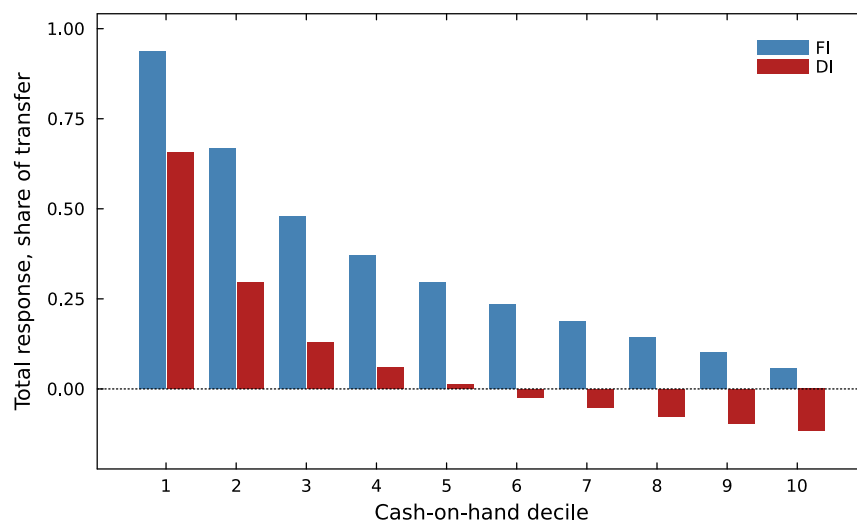
Figure 9 compares the aggregate consumption responses directly. Under full information, the cumulative response over the first two quarters, normalized by the fiscal payment, exceeds 0.7. This is the model's full-information benchmark for the two-quarter general-equilibrium MPC. Under dispersed information, the corresponding response is approximately 0.15 and substantially more gradual. The fiscal payment is therefore much less effective at raising near-term aggregate consumption when households must infer why it was paid.



**Figure 9:** Aggregate consumption response under full and dispersed information

### 5.10.2 Consumption responses across the household distribution

Figure 10 reports the impact consumption response by pre-shock cash-on-hand decile. Responses are divided by the size of the current transfer and can therefore be interpreted as impact consumption responses per unit of fiscal payment.



**Figure 10:** Impact consumption response to a discretionary fiscal innovation

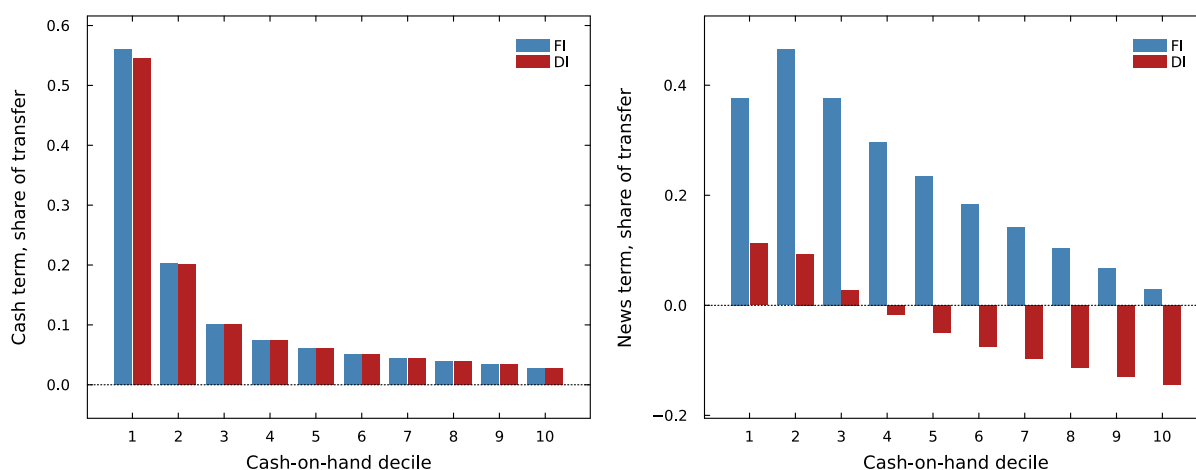
Under full information, the impact response has the standard incomplete-markets shape. It is largest among households with low cash on hand and declines along the distribution.

Under dispersed information, the response is already lower in the bottom deciles. The informational effect becomes increasingly negative higher in the distribution and eventually reverses the sign of the total response.

The direct cash-flow effect is positive under both information structures. The difference arises because a higher transfer also changes beliefs. Under full information, households know that the payment reflects a persistent discretionary fiscal innovation and therefore anticipate additional future transfers. Under dispersed information, the same payment is also evidence that aggregate output may be weak. The resulting downward revision in expected output and labor income offsets the favorable news about future fiscal payments. This negative news channel matters especially for high-cash-on-hand households, whose spending is less tightly tied to current resources.

### 5.10.3 Cash-flow and information decomposition

Figure 11 decomposes the impact response into a cash-in-hand component and a news component. The cash-in-hand component measures the response to the additional current resources while holding beliefs fixed. The news component contains the response to the change in expected future resources and continuation values.



**Figure 11:** Cash-in-hand and news components of the impact consumption response

The cash-in-hand component is almost identical under full and dispersed information. Under full information, the net news component is positive because persistence in  $v_t$  announces additional future transfers. Under dispersed information, the news component is

smaller throughout the distribution and becomes negative for households with sufficiently high cash on hand. These households respond strongly to the negative revision in expected aggregate output, which eventually dominates the favorable news about persistent fiscal support.

Taken together, the results show that the informational content of fiscal policy materially changes both the distributional and aggregate effects of transfers. The direct cash-flow response remains close to that in the full-information economy. What changes is the interpretation of the policy. Because a high transfer can signal weak aggregate conditions, dispersed information induces pessimistic revisions that substantially lower and delay the aggregate consumption response.

## 6 Conclusion

There is suggestive evidence that large stimulus programs based on direct payments to households may induce responses different from those to similarly sized windfall payments at the micro level. This paper addresses this pattern by linking it to evidence on how households form expectations. The main result is that matching such evidence in a model with dispersed information leads to agents rationally inferring bad aggregate conditions from stimulus payments. Stimulus programs are therefore not simply the sum of many windfall gains, but a different economic object.

To tackle the effects of fiscal stimulus programs, this paper has developed a framework for studying macroeconomic dynamics under information frictions that apply to aggregate variables. Households perfectly observe the variables in their budget constraints, but do not observe aggregate states directly. Instead, items of the budget constraint serve as dispersed signals and are used to form rational expectations by applying a Kalman filter to the underlying process or its approximate analogue.

In a basic consumption–savings model, this mechanism generates hump-shaped aggregate consumption responses to aggregate shocks, while responses to idiosyncratic shocks remain front-loaded. MPCs depend on the information carried by shocks, and informational MPCs can be substantially larger than pure MPCs that hold beliefs fixed.

Taking the framework to a general-equilibrium model adds a new mechanism to fiscal policy: A fiscal expansion may reveal private information the government holds, and thus affect household expectations and behavior. The results demonstrate that this can lead to significant heterogeneity in consumption responses and substantially reduce the effectiveness of fiscal stimulus, in line with empirical evidence.

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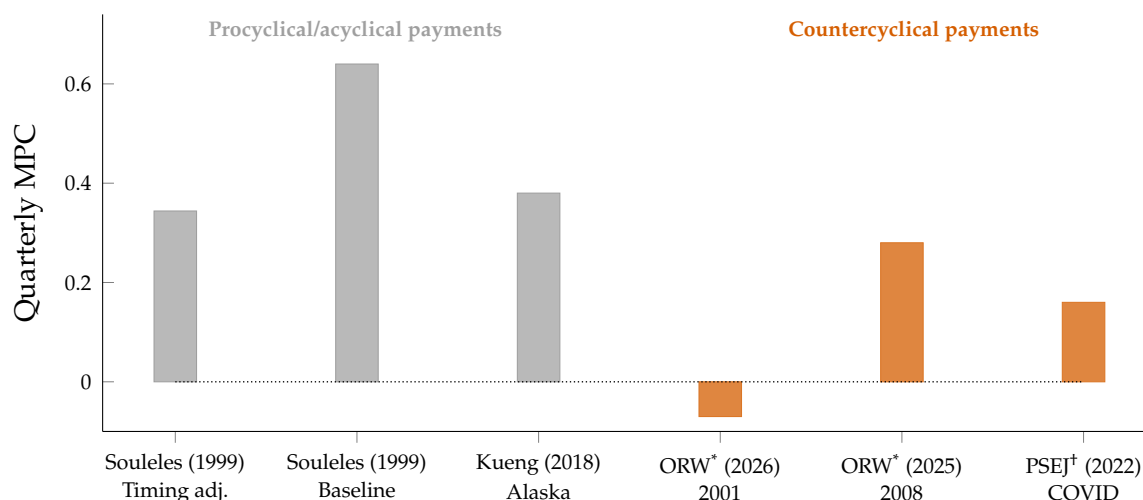
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## A Responses to fiscal transfers: empirical benchmarks

This appendix collects micro- and macro-level consumption benchmarks to complement the empirical evidence and model results.

### A.1 Household MPCs and the policy context

Figure 12 compares selected quarterly MPC estimates from U.S. studies. Responses to regular tax refunds and Alaska dividends exceed the estimates for three countercyclical episodes. Due to differences in settings and underlying data, this comparison is suggestive rather than causal.



**Figure 12:** Selected quarterly household MPC estimates by policy context

*Notes:* Bars show selected point estimates. The Souleles values are alternative specifications from the same study. The Kueng value combines the quarterly responses of nondurable and service spending (0.25) and classified durable purchases (0.13). The negative 2001 point estimate does not establish a significantly negative MPC. \*ORW denotes Orchard, Ramey, and Wieland; †PSEJ denotes Parker, Schild, Erhard, and Johnson. Cyclicity labels refer to the policy context and design, rather than an estimated correlation between payments and output.

Related evidence comes from Goldfayn-Frank et al. (2022), who estimate an overall low MPC of 5.4% across the three countercyclical *Kinderbonus* payments in Germany during COVID. Conversely, Agarwal and Qian (2014) find a large spending response of about 0.80 per dollar from Singapore’s procyclical Growth Dividend. Both studies are excluded because they concern non-U.S. settings and non-quarterly horizons: the *Kinderbonus* estimates use spending within the payment months, while Singapore’s ten-month estimate starts at the announcement, even including spending before disbursement.

The earlier rebates’ lower MPCs are not explained by payment size. Fagereng et al. (2021) document that MPCs decline with lottery prize size. Yet Souleles (1999) reports a mean

household refund of \$874 in 1982–1984 dollars, and Kueng (2018) reports average Alaska dividends of \$1,650 per person, or \$4,600 per household, in 2014 dollars. The 2001 rebates were generally \$300–\$600 per household, and the 2008 payments averaged about \$1,000 per household. In essence, if anything differences in payment size would bias the procyclical or acyclical MPCs downward.

The pattern is noteworthy given that there is evidence of state dependence in MPCs. Sokolova (2023) finds that MPC estimates increase with unemployment rates, while Gross et al. (2020) find higher MPCs out of additional credit availability during recessions and in areas with higher unemployment. The information mechanism pushes the other way: countercyclical payments confer negative news that can offset stronger spending responses associated with weak economic conditions.

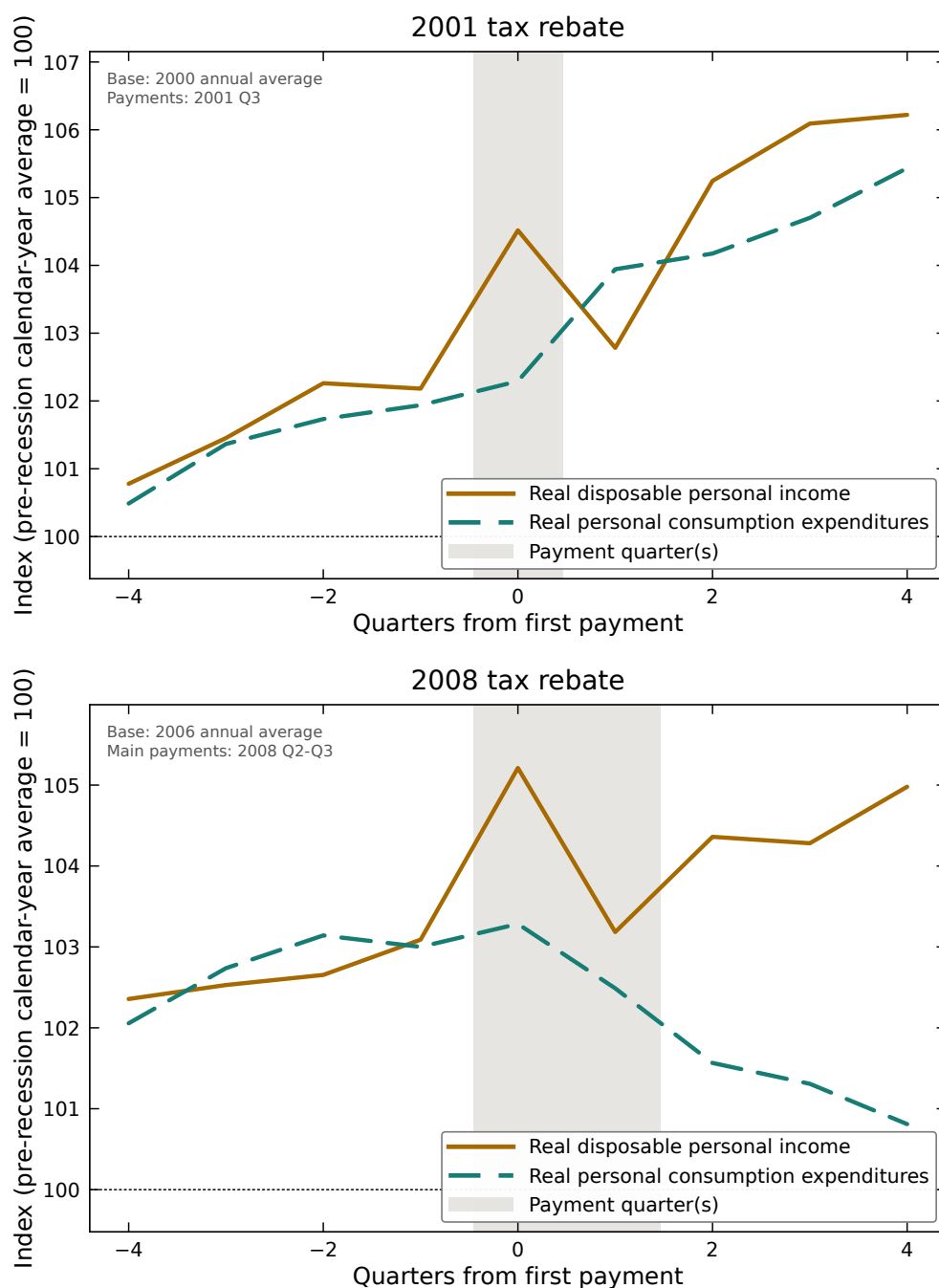
## **A.2 Aggregate income and consumption around stimulus payments**

Figure 13 plots real disposable income and consumption around the 2001 and 2008 U.S. tax rebates. Temporary income spikes do not come with consumption increases. This is suggestive, rather than causal evidence. Nevertheless, these patterns motivate examining why stimulus payments can generate modest aggregate consumption responses, as done by Taylor (2009, 2022), Orchard, Ramey, and Wieland (2025, 2026), and Ramey (2026).

The divergence is even more pronounced during the COVID-19 pandemic (Figure 14). Restrictions, health risks, and other contemporaneous policies directly affected spending, however. The figure illustrates the divergence between aggregate income and consumption. It does not establish that the payments themselves had a small or negative effect on consumption.

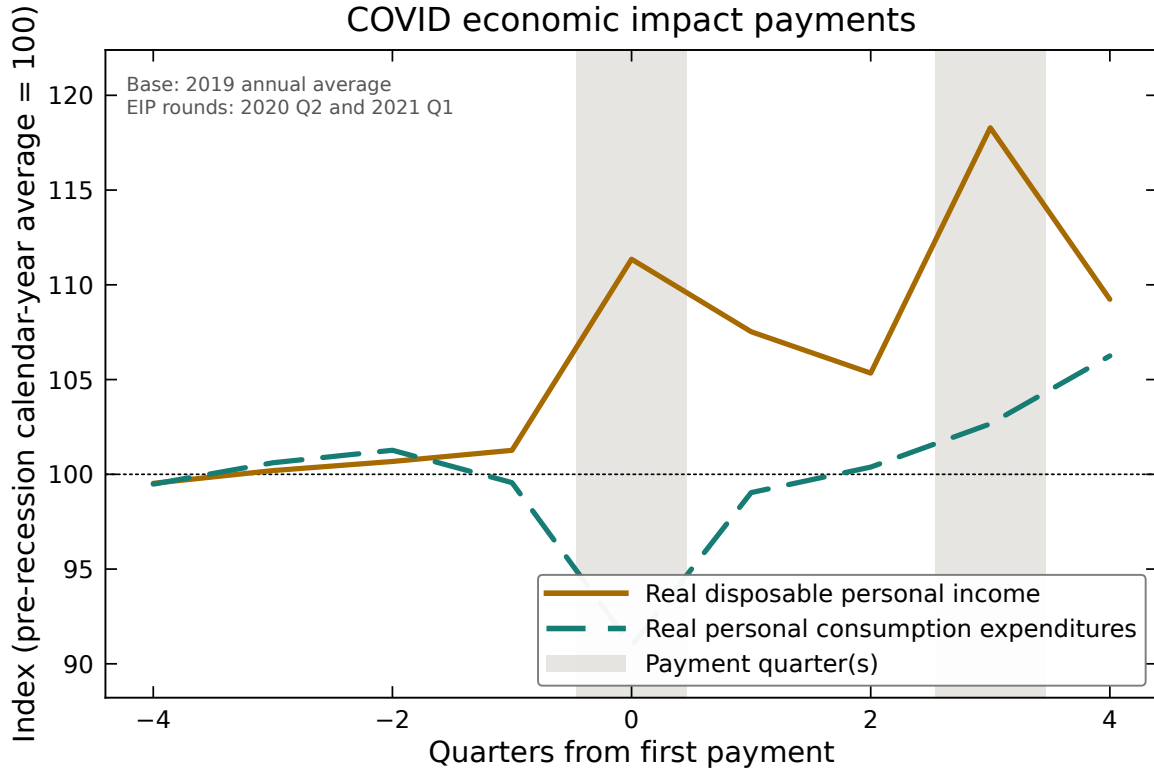
## **B Additional Empirical Results**

This appendix provides additional information on the construction of the SCE and BOP-HH samples and reports supplementary results supporting the empirical analysis in the main text. The experience-effects appendix documents the full specifications behind the SCE core-panel results and the household-income evidence from the BOP-HH. The experimental appendix reports validity checks, income heterogeneity, and the persistence of the fiscal-package treatment effect.



**Figure 13:** Real disposable income and consumption around the 2001 and 2008 tax rebates

*Notes:* Quarterly real disposable personal income (DPIC96, BEA account code A067RX) and real personal consumption expenditures (PCECC96, DPCERX), from the BEA National Income and Product Accounts, retrieved through FRED (U.S. Bureau of Economic Analysis, n.d.-a; U.S. Bureau of Economic Analysis, n.d.-b). The underlying series are seasonally adjusted annual rates in billions of chained 2017 dollars. Each series is indexed to its own calendar-year average: 2000 for the 2001 episode and 2006 for the 2008 episode. Quarter zero is 2001 Q3 and 2008 Q2, respectively. Shading marks payment quarters. Vertical scales differ across panels.



**Figure 14:** Real disposable income and consumption around COVID economic impact payments  
*Notes:* Data sources and units are as in Figure 13 (U.S. Bureau of Economic Analysis, n.d.-a; U.S. Bureau of Economic Analysis, n.d.-b). Each series is indexed to its own 2019 calendar-year average. Quarter zero is 2020 Q2. Shading marks the payment quarters shown in the figure: 2020 Q2 and 2021 Q1.

## B.1 Data and measurement

### B.1.1 SCE core panel

I use SCE core panel between June 2013 and September 2025 to study the relationship between households' financial circumstances and their expectations about aggregate unemployment. The household financial-situation variable is question Q1, which asks whether the household is financially better or worse off than twelve months earlier on a five-point scale from 1, much worse, to 5, much better. Aggregate unemployment expectations are measured by Q4new, the reported percent chance that the US unemployment rate will be higher twelve months later.

The analysis sample retains observations with nonmissing Q1, Q4new, and positive survey weights. The main fixed-effects regressions use SCE survey weights and cluster standard errors by respondent.

For the rationality exercise, I define the forecast as

$$F_{it} = Q4new_{it}/100. \quad (86)$$

The realization is

$$R_{it} = \mathbb{1}\{u_{t+12} > u_t\}, \quad (87)$$

where  $u_t$  is the US unemployment rate in the interview month. The forecast error is

$$FE_{it} = R_{it} - F_{it}. \quad (88)$$

The rationality sample excludes interviews for which the unemployment rate twelve months later is unavailable. The primary covariance estimator clusters by both survey month and respondent. This accounts for the fact that the realization is common to all respondents interviewed in the same month, while forecast and reporting errors may also be correlated within respondents.

### B.1.2 BOP-HH experience panel

The BOP-HH experience sample uses the Scientific Use File (SUF), version 2025Q4 (Deutsche Bundesbank, 2025), and covers waves 4 through 69, corresponding to April 2020 through September 2025. Aggregate beliefs are measured using qualitative questions about whether German unemployment and economic growth will decrease, remain approximately unchanged, or increase over the following twelve months.

Household income is reported in brackets and converted to the midpoint supplied with the scientific-use file. Nominal household-income growth is the log change in this midpoint between consecutive interviews. Real income growth subtracts German HICP inflation over the corresponding interval. Both nominal and real growth are winsorized at their respective first and ninety-ninth percentiles.

The BOP-HH variable refers to total household income and remains defined following a job change or job loss, making it well suited to measuring adverse changes in household resources. One significant issue is that the survey is mostly only available during a time with high (expected) inflation. This together with the measurement of income through brackets or midpoints means many wages increases likely go unmeasured, while movements crossing brackets that are recorded might simply reflect inflation adjustments, possibly even decreases in real wages.

The SCE and BOP-HH aggregate-unemployment variables measure similar but not identical objects. The SCE asks for the probability that unemployment will be higher, while the BOP-HH asks for the direction of change on a qualitative scale. In the shared experience-

effect specifications, belief outcomes are scaled by their full-sample standard deviations.

### **B.1.3 BOP-HH randomized experiment**

The randomized information experiment was conducted in wave 4 of the BOP-HH in April 2020 (Deutsche Bundesbank, 2020). Respondents were assigned to one of six treatment arms:

1. placebo information about European human rights;
2. recession news;
3. uncertainty news;
4. information about the ECB Pandemic Emergency Purchase Programme;
5. information about the German federal fiscal package;
6. government reassurance.

The empirical analysis in the paper compares respondents assigned to the federal fiscal-package treatment, arm 5, with respondents assigned to the placebo treatment, arm 1. Arms 2, 3, 4, and 6 are excluded from the main treatment regressions. The treatment indicator is therefore

$$fp_i = 1\{\text{respondent } i \text{ is assigned to arm 5}\}, \quad (89)$$

within the sample consisting of arms 1 and 5.

Post-treatment GDP expectations are constructed from the eight-bin probability distribution in question 419. The mean forecast uses the bin midpoints and is expressed in percentage points of expected GDP growth. Spending outcomes are stated propensities from 0 to 100 to purchase a house, purchase a car, purchase another large durable good, or increase saving.

All experimental regressions use BOP-HH survey weights and heteroskedasticity-robust standard errors. The standard control set includes gender, age, household size, employment status, and pre-treatment qualitative growth expectations.

**Table 8:** Own financial situation and aggregate unemployment expectations (SCE)

|                                    | Unweighted           | Survey-weighted      |
|------------------------------------|----------------------|----------------------|
| No fixed effects                   | -4.128***<br>(0.121) | -3.943***<br>(0.160) |
| Month fixed effects                | -3.893***<br>(0.122) | -3.759***<br>(0.161) |
| Respondent fixed effects           | -1.858***<br>(0.097) | -1.884***<br>(0.132) |
| Respondent and month fixed effects | -1.577***<br>(0.095) | -1.600***<br>(0.128) |
| Observations                       | 185,086              |                      |

## B.2 Experience effects and forecast efficiency

### B.2.1 Financial conditions and unemployment expectations

The following table reports the full set of specifications relating respondents' household financial situation to their aggregate unemployment expectations. It includes the pooled specification, month fixed effects, respondent fixed effects, and both sets of fixed effects, with weighted and unweighted estimates.

The relationship between household financial conditions and aggregate unemployment expectations is negative across specifications. The month-fixed-effects estimates compare respondents interviewed under the same aggregate conditions, while the respondent-fixed-effects estimates use changes within respondents during their SCE panel spells.

### B.2.2 Forecasts, realizations, and forecast errors

The following table reports the full rationality exercise. In addition to the preferred two-way-clustered standard errors, it includes specifications clustering only by survey month and only by respondent.

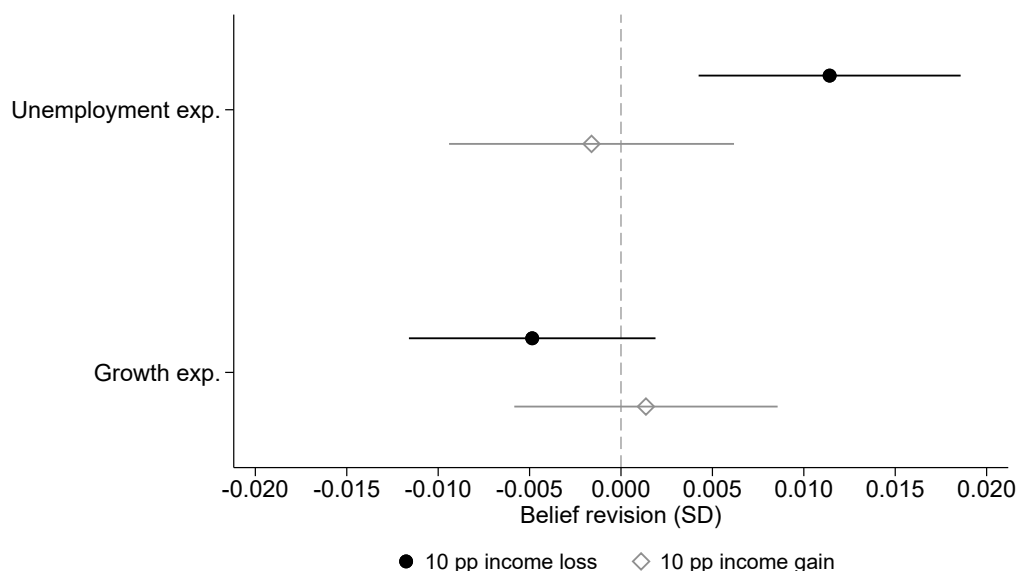
The estimated financial-situation slopes for forecasts and realizations are quantitatively similar. The corresponding forecast-error slope is close to zero and is not statistically distinguishable from zero under two-way clustering. Respondent-only clustering understates uncertainty in the realization equation because the realized aggregate outcome varies only across survey months.

**Table 9:** Forecast efficiency: stated probability versus realized outcome

|                | Coefficient | Two-way SE | Month SE | Respondent SE |
|----------------|-------------|------------|----------|---------------|
| Forecast       | -0.0391***  | (0.0021)   | (0.0016) | (0.0016)      |
| Realization    | -0.0362***  | (0.0105)   | (0.0100) | (0.0034)      |
| Forecast error | 0.0029      | (0.0108)   | (0.0102) | (0.0038)      |
| Observations   |             | 180,863    |          |               |
| Survey months  |             | 144        |          |               |
| Respondents    |             | 24,049     |          |               |

### B.2.3 BOP-HH household-income changes

Figure 15 reports the asymmetric relationship between real household-income changes and revisions in aggregate expectations. Positive and negative income changes enter the same regression with a shared intercept, wave fixed effects, and a control for the number of waves since the previous interview.

**Figure 15:** Real household-income changes and aggregate expectations in the BOP-HH

Source: Research Data and Service Centre (RDSC) of the Deutsche Bundesbank, Bundesbank Online Panel – Households (BOP-HH), April 2020–September 2025, own calculations.

The estimates indicate that revisions in aggregate growth and unemployment expectations are more strongly associated with household-income losses than with household-income gains.

**Table 10:** Randomization balance: placebo versus selected treatment arms

| Covariate                            | Placebo | Policy | Difference | p-value | N   |
|--------------------------------------|---------|--------|------------|---------|-----|
| <i>Panel A: Full analysis sample</i> |         |        |            |         |     |
| Age                                  | 53.358  | 54.088 | 0.730      | 0.557   | 681 |
| Household size                       | 2.248   | 2.232  | -0.015     | 0.854   | 679 |
| Gender                               | 1.510   | 1.444  | -0.066*    | 0.084   | 681 |
| Employed                             | 0.571   | 0.566  | -0.004     | 0.912   | 679 |
| Homeowner                            | 0.683   | 0.641  | -0.042     | 0.246   | 681 |
| Income bracket                       | 2.868   | 2.811  | -0.057     | 0.584   | 647 |
| Prior growth outlook                 | 1.707   | 1.605  | -0.102     | 0.173   | 680 |
| Credit constrained                   | 0.091   | 0.135  | 0.044*     | 0.070   | 680 |
| Joint F-test (Unweighted)            | –       | –      | 1.398      | 0.194   | 645 |
| Joint F-test (Weighted)              | –       | –      | 1.902      | 0.057   | 645 |
| <i>Panel B: Top-income group</i>     |         |        |            |         |     |
| Age                                  | 48.800  | 52.873 | 4.073      | 0.202   | 100 |
| Household size                       | 3.022   | 2.745  | -0.277     | 0.173   | 100 |
| Gender                               | 1.356   | 1.309  | -0.046     | 0.628   | 100 |
| Employed                             | 0.689   | 0.673  | -0.016     | 0.865   | 100 |
| Homeowner                            | 0.822   | 0.764  | -0.059     | 0.475   | 100 |
| Income bracket                       | 5.000   | 5.000  | 0.000      | –       | 100 |
| Prior growth outlook                 | 1.756   | 1.582  | -0.174     | 0.397   | 100 |
| Credit constrained                   | 0.067   | 0.109  | 0.042      | 0.456   | 100 |
| Joint F-test (Unweighted)            | –       | –      | 0.869      | 0.534   | 100 |
| Joint F-test (Weighted)              | –       | –      | 1.062      | 0.395   | 100 |

Source: Research Data and Service Centre (RDSC) of the Deutsche Bundesbank, Bundesbank Online Panel – Households (BOP-HH), April 2020, own calculations.

## B.3 Experimental validity

### B.3.1 Covariate balance

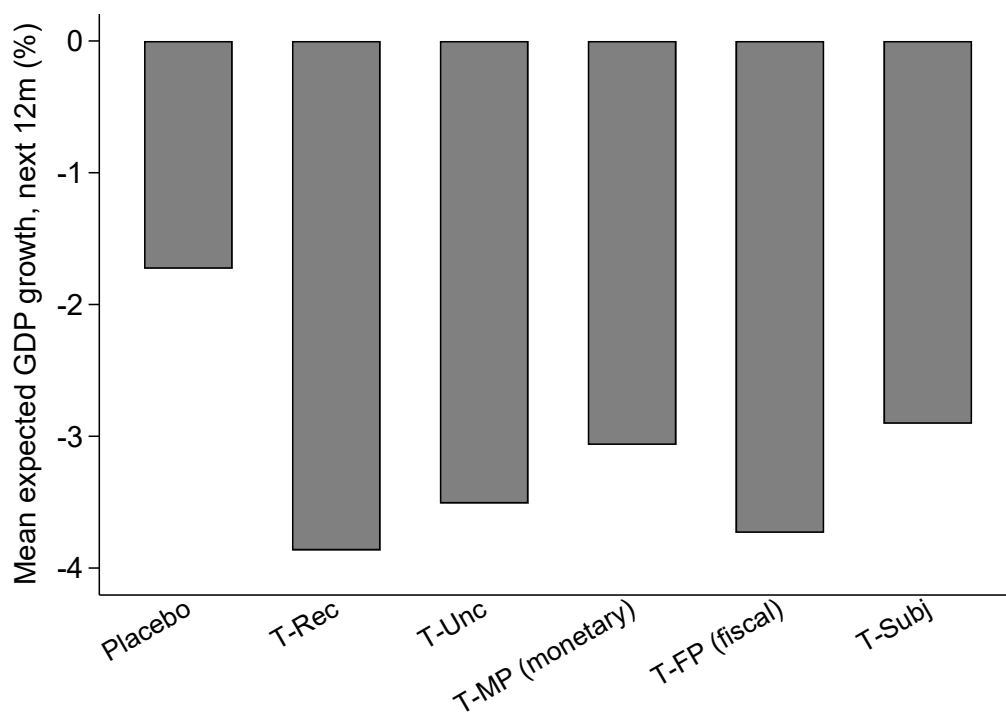
The following table reports balance in pre-treatment covariates between respondents assigned to the fiscal-package treatment and respondents assigned to the placebo treatment.

The unweighted balance tests evaluate whether random assignment generated comparable groups in the arm-5-versus-arm-1 sample. The weighted tests provide a corresponding diagnostic for the weighted estimation sample. Survey weights were not used in treatment assignment, so differences between weighted and unweighted balance tests do not imply a failure of randomization.

### B.3.2 GDP expectations across experimental arms

Figure 16 reports mean post-treatment GDP expectations in all six experimental arms.

Although the main regressions compare only the fiscal-package and placebo arms, the six-arm figure provides a transparent overview of the experimental variation. It also



**Figure 16:** Post-treatment GDP expectations by experimental arm

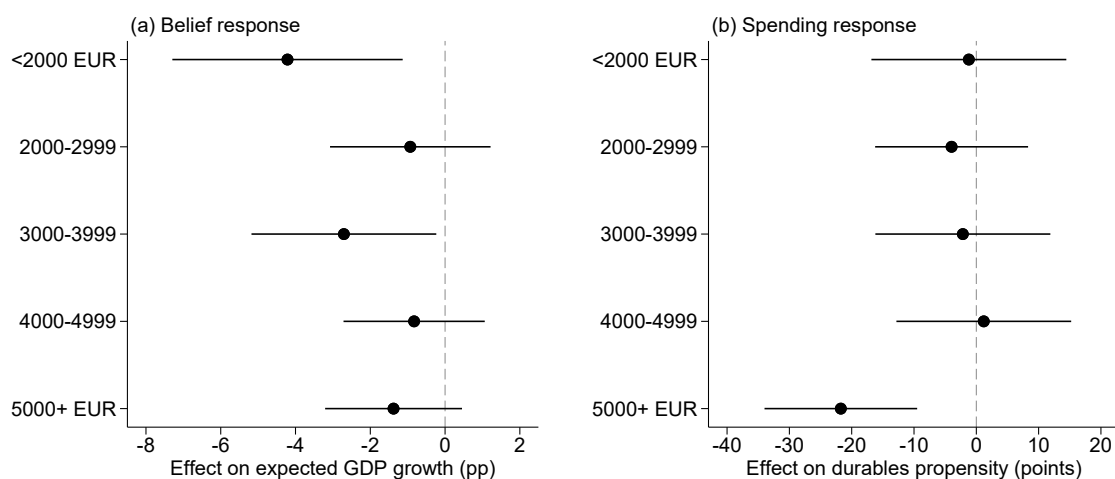
Source: Research Data and Service Centre (RDSC) of the Deutsche Bundesbank, Bundesbank Online Panel – Households (BOP-HH), April 2020, own calculations.

validates the construction of the post-treatment GDP-growth forecast from the probability bins.

## B.4 Spending heterogeneity

### B.4.1 Income-bracket profiles

Figure 17 reports the effects of the fiscal-package treatment on expected GDP growth and intended durable purchases separately across the five household-income groups. Each coefficient compares arm 5 with placebo arm 1.



**Figure 17:** Fiscal-package information, GDP expectations, and intended durable spending by household income

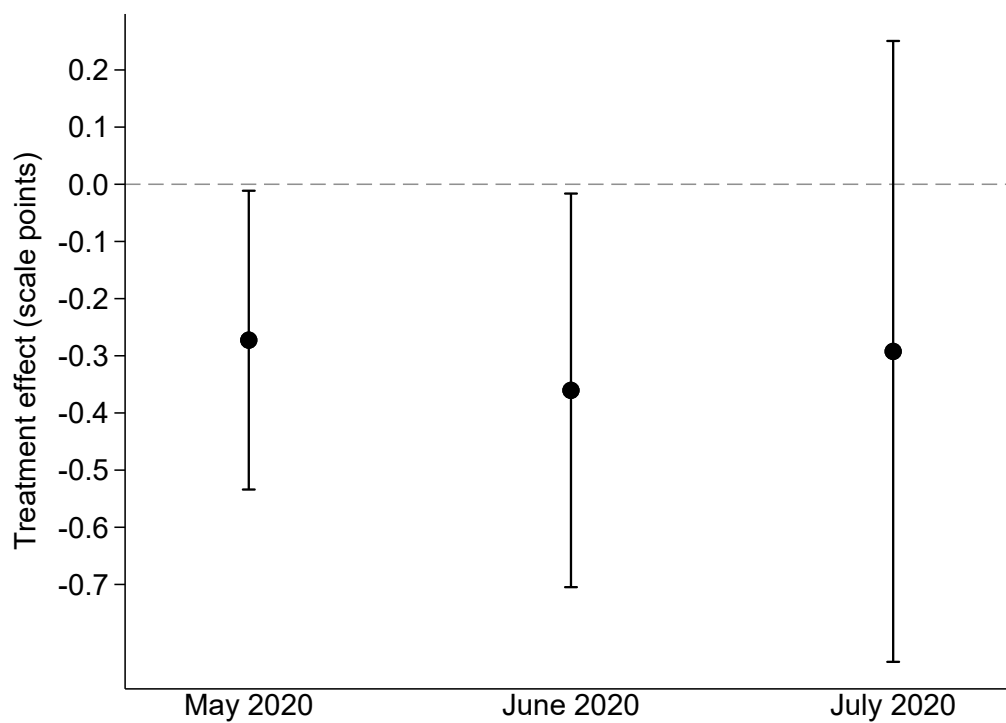
Source: Research Data and Service Centre (RDSC) of the Deutsche Bundesbank, Bundesbank Online Panel – Households (BOP-HH), April 2020, own calculations.

## B.5 Persistence of the fiscal-package belief effect

Figure 18 reports the effect of the fiscal-package information treatment on qualitative growth expectations in the months following the experiment.

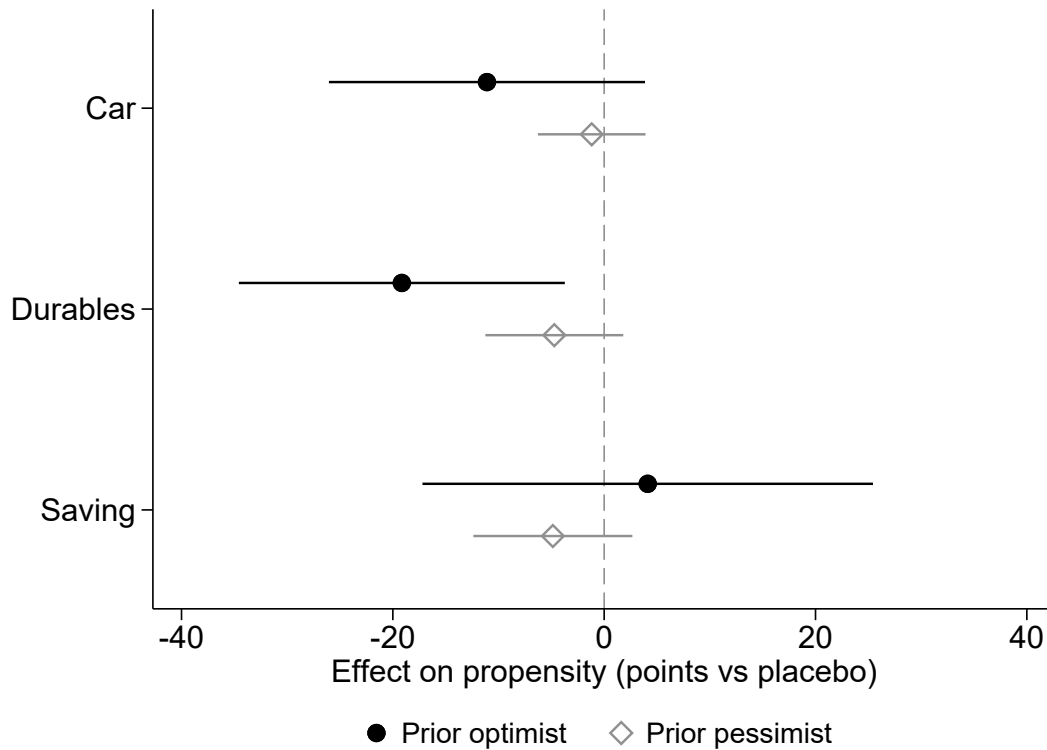
The figure shows whether the initial effect of the fiscal-package treatment remains visible in the subsequent BOP-HH waves. Precision declines over the follow-up period because of rapid panel attrition and because the analysis is restricted to the fiscal-package and placebo arms.

The follow-up outcome is the qualitative five-category growth-expectations question from the regular BOP-HH module. It is not the original post-treatment GDP-density



**Figure 18:** Persistence of the fiscal-package treatment effect on growth expectations

Source: Research Data and Service Centre (RDSC) of the Deutsche Bundesbank, Bundesbank Online Panel – Households (BOP-HH), April–July 2020, own calculations.



**Figure 19:** Spending and saving responses by prior growth expectations

Source: Research Data and Service Centre (RDSC) of the Deutsche Bundesbank, Bundesbank Online Panel – Households (BOP-HH), April 2020, own calculations.

forecast, which was not repeatedly elicited for the surviving experimental households. The persistence exercise therefore examines whether the treatment continues to affect a closely related measure of aggregate growth expectations rather than the exact original outcome. The large standard errors are part due to attrition: Between treatment and control group, there are only 331 respondents left in wave 5, 160 in wave 6 and 111 in wave 7.

## B.6 Prior belief relevance for the effect

Figure 19 reduction in intended spending and the increase in intended saving are concentrated among respondents who were relatively optimistic before receiving the fiscal-package information. For these respondents, the announcement represents more adverse news relative to their prior beliefs.

## C Machine-learning signal-extraction counterfactuals

This appendix describes the alternative forecasting rules used in the ex-post signal-extraction exercise. All methods forecast current aggregate log output  $y_t$  from household-observable variables. Define the current observation vector as

$$o_{it} = (w_{it}, \tilde{T}_t, D_t)'. \quad (90)$$

Here,  $D_t$  is observed exactly through the tax rule. All standardization parameters and forecasting coefficients are estimated using the training sample only. Hyperparameters are selected using expanding-window cross-validation over time blocks.

### C.1 Ridge regression

The Ridge specification uses the most recent  $L$  observations of all three observable series. Define

$$\mathbf{s}_{it} = (o'_{it}, o'_{i,t-1}, \dots, o'_{i,t-L+1})' \in \mathbb{R}^{3L}, \quad L = 12. \quad (91)$$

Equivalently,  $\mathbf{s}_{it}$  contains the current and lagged values of  $w_{it}$ ,  $\tilde{T}_t$ , and  $D_t$ . The inputs are standardized using training-sample means  $m_s$  and standard deviations  $d_s$ :

$$\tilde{\mathbf{s}}_{it} = (\mathbf{s}_{it} - m_s) \oslash d_s. \quad (92)$$

The Ridge forecast is

$$\hat{y}_{it}^{\text{ridge}} = \alpha + \beta' \tilde{\mathbf{s}}_{it}. \quad (93)$$

The parameters solve

$$(\hat{\alpha}, \hat{\beta}) = \arg \min_{\alpha, \beta} \sum_{(i,t) \in \mathcal{T}} (y_t - \alpha - \beta' \tilde{\mathbf{s}}_{it})^2 + n_{\mathcal{T}} \lambda \|\beta\|^2, \quad (94)$$

where  $\mathcal{T}$  denotes the training sample and  $n_{\mathcal{T}}$  is its number of observations. The penalty parameter  $\lambda$  is selected by expanding-window cross-validation.

The current observation vector is an input rather than a predetermined household state. The recursive state required to carry the forecasting rule forward therefore consists of the preceding  $L - 1$  observations of each series, for a total of

$$3(L - 1) = 33 \quad (95)$$

state variables per household.

## C.2 Gated recurrent unit

The GRU replaces the finite observation window with a nonlinear recursive state  $h_{it} \in \mathbb{R}^H$ . Let  $\tilde{o}_{it}$  denote the standardized current observation vector and initialize  $h_{i0} = 0$ . The recursion is

$$r_{it} = \Lambda(W_r \tilde{o}_{it} + U_r h_{i,t-1} + b_r), \quad (96)$$

$$g_{it} = \Lambda(W_g \tilde{o}_{it} + U_g h_{i,t-1} + b_g), \quad (97)$$

$$\tilde{h}_{it} = \tanh(W_h \tilde{o}_{it} + r_{it} \odot (U_h h_{i,t-1}) + b_h), \quad (98)$$

$$h_{it} = (1 - g_{it}) \odot \tilde{h}_{it} + g_{it} \odot h_{i,t-1}. \quad (99)$$

Here,  $\Lambda(v) = 1/(1 + \exp(-v))$  is applied element by element and  $\odot$  denotes elementwise multiplication. The output forecast is

$$\hat{y}_{it}^{\text{GRU}} = \mu_y + \sigma_y (a' h_{it} + c), \quad (100)$$

where  $\mu_y$  and  $\sigma_y$  are computed from the training sample.

The parameters minimize training-sample mean squared error using truncated backpropagation through time. The hidden-state dimension is selected by validation from

$$H \in \{8, 16\}. \quad (101)$$

Validation selects  $H = 16$ . The forecasting rule therefore requires sixteen recursively updated state variables per household.

## C.3 $k$ -nearest neighbors

The  $k$ NN estimator uses the same standardized  $L$ -period vector  $\tilde{\mathbf{s}}_{it}$  as the Ridge specification. Let the reference sample be

$$\mathcal{R} = \{(\tilde{\mathbf{s}}_j, y_j)\}_{j=1}^N, \quad N = 25,000. \quad (102)$$

For a household-period observation  $\tilde{\mathbf{s}}_{it}$ , define the distance to reference observation  $j$  as

$$d_j = \|\tilde{\mathbf{s}}_{it} - \tilde{\mathbf{s}}_j\|_2. \quad (103)$$

Let  $\mathcal{N}_k(\tilde{\mathbf{s}}_{it})$  contain the indices of the  $k$  smallest distances. The inverse-distance weights and output forecast are

$$\omega_j = \frac{1}{d_j + \epsilon_d}, \quad \hat{y}_{it}^{k\text{NN}} = \frac{\sum_{j \in \mathcal{N}_k(\tilde{\mathbf{s}}_{it})} \omega_j y_j}{\sum_{j \in \mathcal{N}_k(\tilde{\mathbf{s}}_{it})} \omega_j}. \quad (104)$$

The constant  $\epsilon_d > 0$  prevents division by zero. The number of neighbors is selected by expanding-window cross-validation, which yields  $k = 160$ .

As with Ridge, the household must retain  $3(L - 1) = 33$  lagged observations. In addition, implementation of the forecasting rule requires storing and searching the reference sample  $\mathcal{R}$ .

## D Design of the fiscal-announcement validation

This appendix describes how households interpret the cumulative fiscal announcement used in Section 5.9. For this validation exercise,  $Y^{\text{ref}}$  is Q1 2020 output. The announcement covers Q2 2020 through Q1 2021 and reveals the normalized four-quarter total

$$s = \sum_{j=0}^3 \tilde{T}_{t+j} = 0.1696. \quad (105)$$

It does not reveal separate quarterly transfers.

### D.1 The announcement as a signal

For this exercise, it is convenient to use the augmented household state

$$q_{it}^a = (\tilde{y}_t, e_{it}, v_t, D_t)'. \quad (106)$$

Let its perceived transition be

$$q_{i,t+1}^a = a_q + A^a q_{it}^a + G^a \epsilon_{i,t+1}. \quad (107)$$

Here  $a_q$ ,  $A^a$ , and  $G^a$  are the augmented counterparts of  $d_q$ ,  $A_q$ ,  $D_q$ , and  $G_q$  in Section 5, obtained by replacing  $y_t$  with  $\tilde{y}_t$  and appending the debt transition. Normalized transfers are selected from this state by

$$\tilde{T}_t = c' q_{it}^a, \quad c = (-\psi_y, 0, 1, 0)'. \quad (108)$$

Under the perceived law, the announced total consists of the observed current transfer, rule-implied future transfers, and planned future fiscal innovations:

$$s = c' q_{it}^a + \alpha_s + \bar{\theta}' q_{it}^a + \sum_{k=1}^3 \omega_k \epsilon_{t+k}^v. \quad (109)$$

The loading on the current state and the deterministic intercept are

$$\bar{\theta}' = \sum_{j=1}^3 c'(A^a)^j, \quad \alpha_s = \sum_{j=1}^3 c' a^{(j)}, \quad (110)$$

where  $a^{(j)}$  is the  $j$ -step transition intercept. The loading of the innovation in quarter  $t + k$  on the cumulative package is

$$\omega_k = \left[ (G^a)' \sum_{j=k}^3 \left( (A^a)^{j-k} \right)' c \right]_{\nu}. \quad (111)$$

The government knows the current state and its own fiscal plan, but not future productivity innovations. Future productivity shocks therefore do not enter (109). Any information about future productivity comes only from propagating the revised belief about the current state.

Numerically, the persistence loadings are

$$\omega = (2.703, 1.896, 0.999). \quad (112)$$

They are close to  $(1 + \rho_{\nu} + \rho_{\nu}^2, 1 + \rho_{\nu}, 1)$ . An innovation at  $t + 1$  affects all three future quarters, an innovation at  $t + 2$  affects two quarters, and an innovation at  $t + 3$  affects only the final quarter. These are persistence loadings, not an exogenous quarterly allocation of the announced package.

Households already know the current transfer. Household  $i$ 's announcement surprise is therefore

$$u_i = s - \tilde{T}_t - \alpha_s - \bar{\theta}' m_{i,t|t}^a = s - E_{it} \left[ \sum_{j=0}^3 \tilde{T}_{t+j} \right], \quad (113)$$

where  $m_{i,t|t}^a$  denotes the posterior mean after observing the current wage, transfer, and tax payment but before receiving the cumulative announcement.

## D.2 Joint Bayesian update

Collect the current state and the three future fiscal innovations in

$$x_i = ((q_{it}^a)', \varepsilon_{t+1}^{\nu}, \varepsilon_{t+2}^{\nu}, \varepsilon_{t+3}^{\nu})'. \quad (114)$$

The prior mean and covariance are

$$E_{it}[x_i] = \begin{pmatrix} m_{i,t|t}^a \\ 0_3 \end{pmatrix}, \quad \text{Var}_{it}(x_i) = \begin{pmatrix} P & 0 \\ 0 & \sigma_{\nu}^2 I_3 \end{pmatrix}. \quad (115)$$

The future fiscal innovations are independent before the announcement. Define

$$S = \bar{\theta}' P \bar{\theta} + \sigma_v^2 \omega' \omega. \quad (116)$$

The posterior means are

$$E_{it}[q_{it}^a | s] = m_{i,t|t}^a + \frac{P \bar{\theta}}{S} u_i, \quad E_{it}[\varepsilon_{t+k}^v | s] = \frac{\sigma_v^2 \omega_k}{S} u_i. \quad (117)$$

After the current signals, uncertainty about the state is rank one:

$$P = p_y d d', \quad d = (1, -1, \psi_y, 0)'. \quad (118)$$

The current transfer is constant along this remaining uncertainty direction because  $c'd = 0$ . Let

$$\lambda = \bar{\theta}' d = -0.0635. \quad (119)$$

The signal variance becomes

$$S = p_y \lambda^2 + \sigma_v^2 \omega' \omega = 2.2 \times 10^{-6} + 1.54 \times 10^{-4} = 1.56 \times 10^{-4}. \quad (120)$$

The share of the announcement surprise attributed to uncertainty about the current state is

$$\varphi = \frac{p_y \lambda^2}{S} = 0.014, \quad 1 - \varphi = 0.986. \quad (121)$$

Thus, households attribute 1.4% of the surprise to worse current conditions and 98.6% to the fiscal plan.

The output-belief update is

$$\Delta \hat{y}_{it} = \frac{p_y \lambda}{S} u_i = -0.219 u_i. \quad (122)$$

The posterior gains on the future fiscal innovations are

$$(E_{it}[\varepsilon_{t+1}^v | s], E_{it}[\varepsilon_{t+2}^v | s], E_{it}[\varepsilon_{t+3}^v | s]) = (0.224, 0.157, 0.083) u_i. \quad (123)$$

The inferred plan is front-loaded because an early innovation is more effective at generating a large cumulative package. Equivalently, the household selects the least statistically surprising sequence of future fiscal innovations consistent with the announced total. The contributions to the cumulative surprise are:

The contribution assigned to the  $t + 1$  innovation includes its persistent effects on transfers in  $t + 1$ ,  $t + 2$ , and  $t + 3$ . It should not be interpreted as the share paid in quarter  $t + 1$ .

**Table 11:** Posterior decomposition of the announcement surprise

| Explanation                                              | Share of announcement surprise |
|----------------------------------------------------------|--------------------------------|
| Worse current economic state                             | 1.4%                           |
| Planned innovation at $t + 1$ and its persistent effects | 60.5%                          |
| Planned innovation at $t + 2$ and its persistent effects | 29.8%                          |
| Planned innovation at $t + 3$                            | 8.3%                           |

### D.3 Mapping the update into GDP expectations

The model counterpart of the survey outcome is average expected output over the four-quarter window relative to Q1 2020:

$$G_i = \frac{100}{4} \sum_{j=0}^3 E_{it}[\tilde{y}_{t+j} \mid s]. \quad (124)$$

The treatment effect is linear in the household-specific surprise:

$$\Delta G_i = 100 (\Gamma_{\text{state}} + \Gamma_{\text{plan}}) u_i. \quad (125)$$

The current-state component is

$$\Gamma_{\text{state}} = \frac{1}{4} \sum_{j=0}^3 \left[ (A^a)^j \frac{P\bar{\theta}}{S} \right]_1 = -0.20704, \quad (126)$$

and the fiscal-plan component is

$$\Gamma_{\text{plan}} = \frac{1}{4} \sum_{j=1}^3 \sum_{k=1}^j \left[ (A^a)^{j-k} G_v^a \right]_1 \frac{\sigma_v^2 \omega_k}{S} = 0.00292. \quad (127)$$

Consequently,

$$\Delta G_i = -20.4118 u_i. \quad (128)$$

At the mean model surprise,  $\bar{u} = 0.11199$ , the model predicts

$$\overline{\Delta G} = -20.4118 \times 0.11199 = -2.286 \text{ percentage points}. \quad (129)$$

The empirical effect of  $-1.755$  percentage points corresponds to a surprise of approximately  $u = 0.086$ . At this surprise,

$$\Delta \hat{\tilde{y}}_{it} = -0.219 \times 0.086 = -0.019, \quad (130)$$

and

$$\Delta G_{\text{state}} \approx -1.79 \text{ pp}, \quad \Delta G_{\text{plan}} \approx 0.03 \text{ pp}, \quad \Delta G \approx -1.76 \text{ pp}. \quad (131)$$

The bad-state inference lowers expected output in every quarter because the revision to current output is persistent. This does not represent news about additional future productivity shocks. The plan component raises expected output, but only slightly because fiscal innovations enter the perceived output law with a small coefficient. Under full information, the current state is already known, so the state component is absent and the announcement produces only the positive plan effect.

## D.4 Conditional fiscal paths and consumption

The announcement fixes only one cumulative total. It does not identify the three future fiscal innovations separately. Let  $\Sigma_x$  denote the prior covariance of  $x_i$  and let  $g$  collect the corresponding signal loadings. The posterior covariance is

$$\text{Var}_{it}(x_i | s) = \Sigma_x - \frac{\Sigma_x g g' \Sigma_x}{g' \Sigma_x g}. \quad (132)$$

The individual quarterly innovations therefore remain uncertain and become negatively correlated after conditioning on the fixed total.

For a prior draw  $x_i^0$ , the simulation constructs a treated draw as

$$x_i^T = E_{it}[x_i | s] + x_i^0 - \frac{\Sigma_x g}{g' \Sigma_x g} g' x_i^0. \quad (133)$$

Every treated draw can imply a different quarterly fiscal path, but every draw satisfies the same announced cumulative package. The household propagates the conditional path forward, observes future wages, transfers, and taxes, updates its beliefs, and applies the stored consumption policy.

The plan component is quantitatively important for expected future transfer income even though it has little effect on expected output. There is, however, one implementation qualification. The stored consumption policy does not include the remaining announced fiscal plan as an explicit household state variable. Future transfers affect consumption when they arrive and through subsequent filtering, but the current policy does not fully incorporate anticipatory consumption responses to the entire announced path. The spending comparison in the main text should therefore be interpreted as a directional validation rather than a complete quantitative implementation of forward-looking spending responses.